

WHAT MATTERS FOR REPRESENTATION ALIGNMENT: GLOBAL INFORMATION OR SPATIAL STRUCTURE?

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ABSTRACT

Representation alignment (REPA) accelerates generative training by distilling features from a pretrained vision encoder into intermediate diffusion representations. We study a central question for this paradigm: which property of the target representation is most predictive for generation, its *global* semantic information (e.g., measured by ImageNet-1K accuracy) or its spatial structure (i.e. pairwise cosine similarity between patch tokens)? Although current practice often treats stronger global semantic performance as evidence of a better alignment target, our empirical study across 27 vision encoders and multiple diffusion-model scales shows a different pattern: spatial structure in the target representation is a substantially stronger predictor of generation quality under REPA than global discriminative performance. Motivated by this observation, we introduce two minimal changes that make the alignment objective transfer *spatial* information more directly from the external representation to diffusion features. We replace the standard MLP projection layer in REPA with a lightweight convolutional projection and apply spatial normalization to the external representation. The resulting method, iREPA, requires <4 lines of code and consistently improves the convergence speed of REPA across diverse vision encoders, model sizes, and training variants (including REPA, REPA-E, Meanflow, JiT etc). Together, these results reposition representation alignment around spatial feature geometry and provide a simple, empirically supported route for improving the training efficiency of generative models.

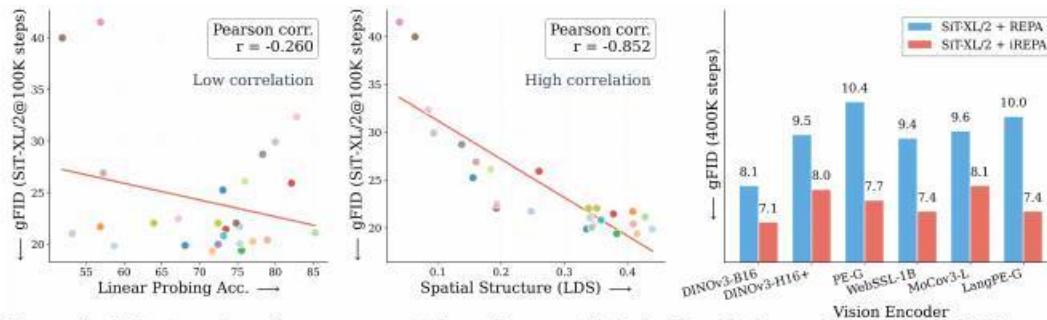


Figure 1: **What matters for representation alignment?** *Left:* Correlation analysis across 27 diverse vision encoders. Surprisingly, contrary to the prevailing wisdom, we find that spatial structure, rather than global performance (measured by linear probing accuracy), drives the generation performance of a target representation. *Right:* We further study this by introducing two simple modifications to accentuate the transfer of spatial features from target representation to diffusion model. Our simple approach consistently improves the convergence speed of REPA across diverse settings.

1 INTRODUCTION

Representation alignment has emerged as an effective mechanism for accelerating the training of diffusion transformers (Ma et al., 2024; Peebles & Xie, 2023). By constraining intermediate diffusion features to match representations from pretrained self-supervised vision encoders, recent methods improve both convergence speed and generation quality (Yu et al., 2024; Leng et al., 2025a). The



Figure 2: **Motivating examples — spatial structure matters.** Metrics comparison showing inverse relationship between ImageNet accuracy and generation quality. *Left:* PE-G, despite having significantly higher validation accuracy (82.8% vs. 53.1%), shows worse generation quality compared to SpatialPE-B (Bolya et al., 2025). *Right:* Similarly, WebSSL-1B (Fan et al., 2025) also shows much better global performance (76.0% vs. 53.1%), but worse generation. **Spatial Self-Similarity:** We find that spatial structure instead provides a better predictor of generation quality than global performance. See §3 for spatial structure metric. All results reported at 100K using SiT-XL/2 and REPA.

representational property responsible for these gains, however, remains insufficiently characterized. The central question is whether alignment is effective primarily because it transfers stronger global semantic information, typically assessed through linear probing performance, or because it transfers spatial structure, encoded in the relational geometry among patch-token representations.

Resolving this question directly affects how target representations should be selected, evaluated, and modified for generative training. A prevalent assumption is that the usefulness of an encoder for representation alignment should track ImageNet-1K validation accuracy, a standard proxy for global semantic understanding (Oquab et al., 2024; Chen et al., 2021). Under this interpretation, target representations with stronger ImageNet performance should also yield better generation when used for REPA (Yu et al., 2024). Consequently, increases in the linear probing accuracy of diffusion features are often taken as evidence that representation alignment operates by improving global semantic content. The following quote from REPA (Yu et al., 2024) summarizes this prevailing interpretation:

“When a diffusion transformer is aligned with a pretrained encoder that offers more semantically meaningful representations (i.e., better linear probing results), the model not only captures better semantics but also exhibits enhanced generation performance, as reflected by improved validation accuracy with linear probing and lower FID scores.”

This paper tests that conventional account by comparing the explanatory power of global semantic information and spatial structure for representation alignment. We conduct a large-scale empirical analysis over diverse vision encoders, including recent large vision foundation models such as WebSSL (Fan et al., 2025), DINOv3 (Siméoni et al., 2025), perceptual encoders (Bolya et al., 2025), and C-RADIO (Heinrich et al., 2024), and identify three findings that clarify what makes an external representation effective for generation.

Higher validation accuracy does not imply a better representation for generation. Contrary to the semantic-strength hypothesis, higher global semantic performance, measured by ImageNet-1K linear probing accuracy, is not a reliable indicator of a better target for generation. For example, PE-Spatial-B is a spatially tuned model derived from PE-Core-G (Bolya et al., 2025). Although PE-Spatial-B has substantially lower validation accuracy¹ on patch tokens than PE-Core-G (53.1% vs. 82.8%), it yields better generation with REPA (Figure 2). Similarly, WebSSL-1B (Fan et al., 2025) has much stronger global performance (76.0% vs. 53.1%) but worse generation. In Section §2, we show that this pattern recurs across target representations, indicating that global semantic accuracy alone is an inadequate criterion for selecting alignment targets.

Spatial structure is substantially more predictive of generation performance than global information. To quantify this relationship, we introduce straightforward metrics for the spatial self-similarity structure (Shechtman & Irani, 2007) among patch tokens (§3). We then measure correlations among generation FID under REPA, linear probing accuracy, and spatial structure across 27 vision encoders and three model sizes (SiT-B, SiT-L, SiT-XL). All spatial-structure metrics exhibit

¹Similar to REPA (Yu et al., 2024), we use linear probing accuracy on patch tokens to measure global semantic performance of external representation as only patch tokens are used for representation alignment.

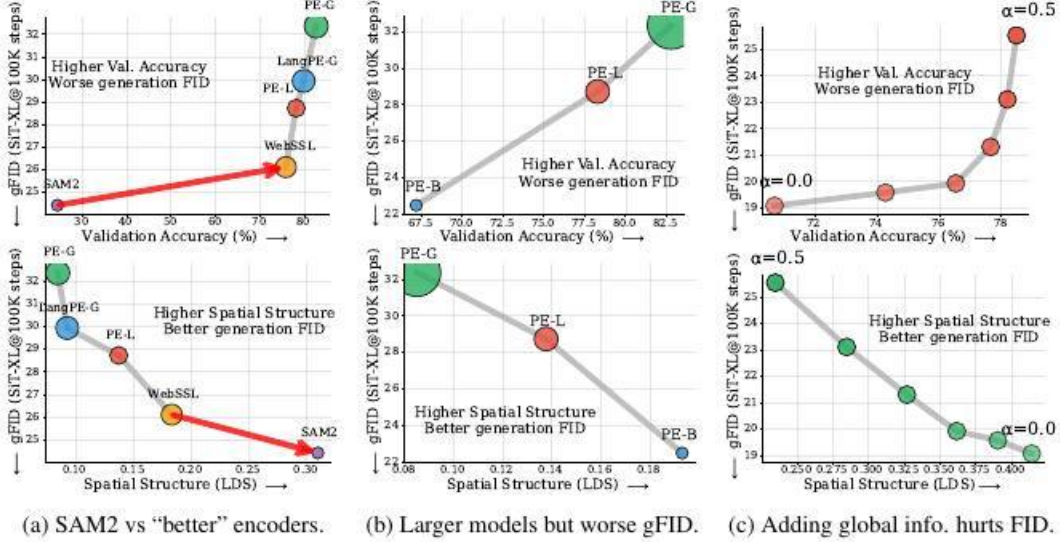


Figure 3: **Higher global information does not imply better REPA performance.** **Top row:** Several trends showing global performance does not correlate well with generation FID when using REPA. (a) SAM2-S with only validation accuracy of 24.7% results in better generation performance with REPA compared with other models with $\sim 60\%$ higher validation accuracy. (b) Larger encoders within same family can have better validation but worse generation performance. (c) Adding global information to patch tokens via CLS token improves global performance but hurts generation. **Bottom row:** We show that spatial structure rather than global performance provides a better indicator for generation. Please see §3 for large-scale detailed analysis across different spatial structure metrics. All results are reported w/o classifier free guidance, SiT-XL/2 w/ REPA and 250 NFE (Yu et al., 2024) for inference.

substantially stronger correlation with generation FID (Pearson $|r| > 0.852$) than ImageNet-1K validation accuracy ($|r| = 0.26$). Additional experiments support the same interpretation: external representations with limited global information can still provide significant gains with REPA when they preserve useful spatial organization. SAM2, for instance, yields better generation with REPA than encoders whose ImageNet accuracy is $\sim 60\%$ higher (Fig. 3). Likewise, although they are not the strongest targets, classical spatial features such as SIFT (Lowe, 1999) and HOG (Dalal & Triggs, 2005) still produce meaningful gains with representation alignment (§3).

Accentuating spatial features improves convergence performance. We operationalize this analysis through two simple modifications (< 4 lines of code) that strengthen the transfer of spatial information from the target representation to the diffusion model. First, we introduce a spatial regularization layer that increases spatial contrast in the target representation. Second, we show that the standard MLP projection layer, which maps diffusion features into the target-representation dimension, attenuates spatial information (Figure 6a); replacing it with a simple convolutional layer preserves this structure more effectively. The resulting method, iREPA, consistently improves the convergence speed of REPA across variations in encoders, model sizes, and training recipes, including REPA-E (Leng et al., 2025a) and Meanflow (Geng et al., 2025) w/ REPA.

The paper makes three connected contributions:

- We provide a large-scale empirical analysis showing that spatial structure is a substantially stronger predictor of REPA effectiveness for diffusion training than global semantic performance.
- We introduce straightforward and fast-to-compute Spatial Structure Metrics (SSM) whose correlations with downstream FID are markedly stronger than those of linear probing scores.
- We propose iREPA, a minimal modification of REPA that accentuates spatial-information transfer and consistently improves convergence speed across target representations, model architectures, and training variants (REPA, REPA-E, Meanflow w/ REPA, JiT w/ REPA etc).

2 MOTIVATION: GLOBAL PERFORMANCE IS NOT SUFFICIENT

We first present motivating evidence that higher global performance, measured by ImageNet-1K accuracy, does not by itself identify a better target representation for REPA. In §3, we show that these

otherwise puzzling observations are explained more consistently by measuring the spatial structure of the target representation.

Recent vision encoders. We compare several recent vision encoders, including Perceptual Encoders (Bolya et al., 2025), WebSSL (Fan et al., 2025), and DINOv3 (Siméoni et al., 2025). Consider PE-Spatial-B (80M), a small spatially tuned model derived from PE-Core-G (1.88B) (Bolya et al., 2025). As shown in Figure 2, PE-Core-G achieves much higher ImageNet-1K accuracy (82.8% vs 53.1%) but performs worse when used as the target representation for REPA (FID 32.3 vs 21.0). Similarly, WebSSL-1B (1.2B) achieves much higher ImageNet-1K accuracy (76.0% vs 53.1%) but also performs worse when used as the target representation for REPA (FID 26.1 vs 21.0).

SAM2 outperforms vision encoders with much higher ImageNet-1K accuracy. To examine the effect of limited global information, we analyze the SAM2-S vision encoder (46M) (Ravi et al., 2024), a small model with validation accuracy of only 24.1% (Fig. 3a). When used for REPA, SAM2-S nevertheless achieves better FID than vision encoders with substantially higher ImageNet-1K accuracy, e.g., PE-Core-G (82.8%).

Larger models within same encoder family can have similar or worse generation performance. A common expectation is that larger variants within an encoder family provide better representations, as reflected by ImageNet-1K accuracy. For representation alignment, however, larger variants often lead to similar (DINOv2) or even *worse* (PE, Cradio) generation performance (Fig. 3b). Notably, (Yu et al., 2024) report a similar observation for DINOv2 and explain it as “we hypothesize is due to all DINOv2 models being distilled from the DINOv2-g model and thus sharing similar representations”. We later show that this trend is better accounted for by spatial structure (§3).

Adding more global information can hurt generation. To test whether additional global information benefits representation alignment, we conduct controlled experiments that inject global semantics from the CLS token into local patch tokens, using DINOv2 as the encoder. The CLS token mixing operation is $\mathbf{p}_i^{\text{new}} = \mathbf{p}_i + \alpha \cdot \mathbf{c}$, where $\mathbf{p}_i$ denotes the i^{th} patch token, $\mathbf{c}$ the CLS token, and $\alpha \in [0, 0.5]$ controls the mixing strength. As shown in Figure 3c, increasing α from 0 to 0.5 monotonically improves linear probing accuracy from 70.7% to 78.5%. Generation quality moves in the opposite direction: FID worsens from 19.2 at $\alpha = 0$ to 25.4 at $\alpha = 0.5$.

Together, these observations show that the global performance of a representation is not a sufficient indicator of its REPA performance. We next show that spatial structure provides a more informative signal (§3).

- **Finding 1.** *Better global semantic information (e.g., ImageNet-1K accuracy) does not imply a better representation for generation.*

3 SPATIAL STRUCTURE BETTER PREDICTS REPA PERFORMANCE

We next test whether spatial structure provides a stronger empirical account of REPA performance than global semantic information. To make this comparison quantitative, we consider several straightforward metrics that measure the spatial self-similarity structure (Shechtman & Irani, 2007) of target representations. Across encoders and model scales, these spatial metrics correlate substantially more strongly with gFID than ImageNet-1K accuracy, and they also account for the previously unexplained observations in §2.

Measuring spatial self-similarity structure. Given an image $\mathcal{I}$ and vision encoder $\mathcal{E}$, we define:

- *Self-similarity.* Let $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ denote the extracted patch representations, with $T = H \times W$ patches. Let the kernel $K_{\mathcal{X}}(\cdot, \cdot) \in \mathbb{R}$ measure appearance self-similarity (Shechtman & Irani, 2007) between patch tokens. We use the cosine kernel $K_{\mathcal{X}}(t, t') = \frac{\langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2}$.
- *Spatial distance.* Let $\mathcal{P} \in \mathbb{N}^{T \times 2}$ denote the spatial location of each of the T tokens in coordinate space, and let $d(\cdot, \cdot) \in \mathbb{N}$ denote the Manhattan distance between token pairs.

We define a *spatial* self-similarity metric by measuring how self-similarity $K_{\mathcal{X}}$ varies with lattice distance d between patch tokens. Larger values indicate stronger spatial organization, meaning that nearby patches are more similar to one another than distant patches. By default, we use a simple

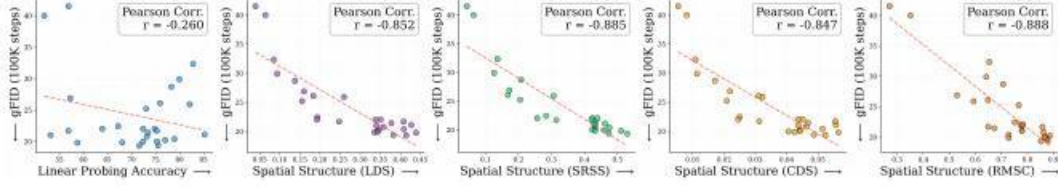


Figure 4: **Spatial structure shows higher correlation with generation quality than linear probing.** Correlation analysis across 27 diverse vision encoders, SiT-XL/2 and REPA. Linear probing shows weak correlation with FID (Pearson $|r| = 0.260$), while all spatial structure metrics: LDS ($|r| = 0.852$), SRSS ($|r| = 0.885$), CDS ($|r| = 0.847$), and RMSC ($|r| = 0.888$), demonstrate much stronger correlation with generation performance. See Fig. 10 for detailed plots with encoder labels.

correlogram-contrast metric comparing local and distant token pairs (Huang et al., 1997):

$$\text{LDS}(\mathcal{X}, \mathcal{P}) = \mathbb{E}_{d(t,t') \in (0, r_{\text{near}})} K_{\mathcal{X}}(t, t') - \mathbb{E}_{d(t,t') \geq r_{\text{far}}} K_{\mathcal{X}}(t, t').$$

The final spatial self-similarity score (LDS) is computed as the expectation over patch representation $\mathcal{X}$. We use $r_{\text{near}} = r_{\text{far}} = H/2$ here, and we find that the observed correlations are robust to the exact choices of these thresholds. Appendix B provides the exact details and alternative spatial metrics (CDS, SRSS, RMSC), which exhibit the same empirical behavior (see Fig. 4).

Spatial structure correlates more strongly with generation performance than linear probing. We perform a large-scale correlation analysis across 27 diverse vision encoders. As shown in Figure 4, the commonly used linear probing score has weak correlation with generation performance (Pearson $|r| = 0.26$), whereas all SSM metrics show much stronger correlation (Pearson $|r| > 0.85$).

Generalization across model scales. We verify the same relationship across different model scales (SiT-B, SiT-L, SiT-XL) in Figure 5. Linear probing remains weakly correlated across scales ($|r| < 0.306$), while spatial structure remains strongly correlated with generation performance ($|r| > 0.826$).

- **Finding 2.** *Spatial structure correlates much more strongly with generation performance than linear probing. We next use spatial metrics to explain previously unexplained trends.*

Spatial structure can explain previously unexplained trends. As discussed in §2, global performance (ImageNet-1K accuracy) is not a reliable predictor of effectiveness for representation alignment. The spatial metrics above provide a more consistent explanation of those results.

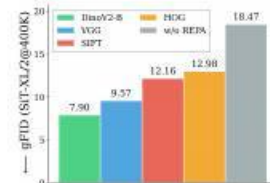
(1) *PE-Spatial-B vs PE-Core-G:* Figure 2 shows that PE-Core-G achieves much higher ImageNet-1K accuracy (82.8% vs 53.1%) but performs worse when used as the target representation for REPA (FID 32.3 vs 22.0). The spatial-structure metric resolves this discrepancy: despite its lower global performance, PE-Spatial-B exhibits stronger spatial structure than PE-Core-G, matching its better generation performance.

(2) *SAM2 outperforms “better” vision encoders:* §2 shows that although SAM2 achieves much lower ImageNet-1K accuracy (24.1%), it leads to better generation than encoders with $\sim 60\%$ higher accuracy. As shown in Fig. 3a, this advantage is consistent with SAM2’s stronger spatial structure.

(3) *Larger models in same encoder family underperform:* As shown in Fig. 3b, larger models in the same encoder family can improve ImageNet-1K accuracy while degrading spatial structure, leading to worse generation performance with REPA. This pattern is consistent with observations from Yu et al. (2024), who find that larger models can have similar or worse generation performance.

(4) *Adding global information to patch tokens via CLS token hurts generation:* In Figure 3c, increasing global information by mixing the CLS token with patch tokens improves global performance. However, CLS mixing reduces spatial contrast among patch tokens, causing tokens from unrelated regions (e.g., foreground object and background) to exhibit high similarity. The resulting reduction in spatial structure corresponds to worse generation performance.

If spatial structure matters more, can we use SIFT or HOG features for REPA? Yes, although they are not the strongest targets. Classical spatial features such as SIFT (Lowe, 1999), HOG (Dalal & Triggs, 2005), and intermediate VGG features (Simonyan & Zisserman, 2014)



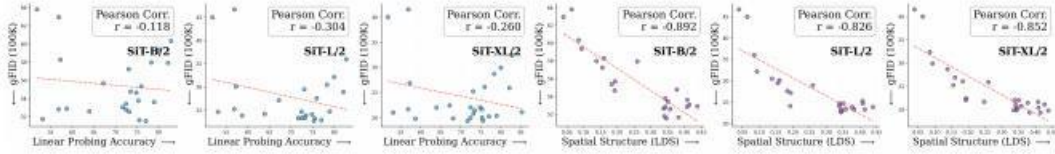


Figure 5: **Correlation analysis across model scales.** Across different model scales, we find that spatial structure (right) consistently shows higher correlation with gFID than linear probing (left).

all produce performance gains with REPA. This result further indicates that representation alignment can benefit from spatial features even without strong global semantic information.

Can spatial metrics explain gains with REPA? Yes. The introduced spatial metrics explain both the gains obtained by REPA and the improvements from our revised training recipe, iREPA, which we introduce next in §4.



4 iREPA: IMPROVING REPRESENTATION ALIGNMENT BY ACCENTUATING SPATIAL STRUCTURE

To further investigate the role of spatial structure in representation alignment, we introduce two straightforward modifications to the original REPA training recipe. Both modifications are designed to enhance the transfer of spatial features from the teacher vision encoder to the student diffusion transformer while leaving the overall alignment framework unchanged.

Convolutional projection layer instead of MLP. The standard REPA approach uses a 3-layer MLP projection to map diffusion features into the dimensionality of the external representation. As shown in Figure 6a, this projection is lossy with respect to spatial contrast between patch tokens. We therefore replace the MLP with a lightweight convolutional layer (kernel size 3, padding 1) that operates directly on the spatial grid. The convolutional projection preserves local spatial relationships through its locality bias while serving the same dimensional-alignment role as the original MLP.

Spatial normalization layer. Similar to Siméoni et al. (2025), we find that patch tokens of pretrained vision encoders contain a substantial global component, as indicated by the high linear probing scores obtained from the mean of patch tokens (Figure 14). Figure 6b further shows that mixing this global information (the mean of patch tokens) into the patch tokens improves global performance but reduces spatial contrast between individual tokens. As a result, tokens from otherwise unrelated regions (e.g., foreground object and background) can exhibit high similarity.

Motivated by the results in §3, we reduce this global component in order to increase spatial contrast among patch tokens. The intended effect is to strengthen the pairwise spatial signal provided to the REPA objective. To this end, we add a simple spatial normalization layer (Ulyanov et al., 2016) to the patch tokens of the target representation:

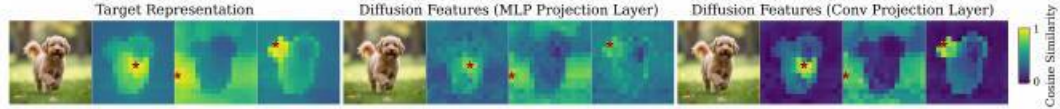
$$\mathbf{y} = \frac{\mathbf{x} - \gamma \mathbb{E}[\mathbf{x}]}{\sqrt{\text{Var}[\mathbf{x}] + \epsilon}},$$

where $\mathbf{x} \in \mathbb{R}^{B \times T \times D}$ represents the patch tokens, the expectation and variance are computed across the spatial dimension, and $\epsilon = 10^{-6}$ ensures numerical stability.

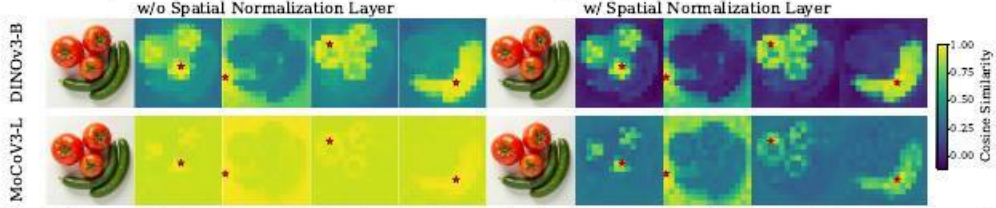
Algorithm 1 Summary of key iREPA modifications to standard REPA training

```
# 1. Conv projection instead of MLP
proj_layer = nn.Conv2d(D_in, D_out, kernel_size=3, padding=1)

# 2. Spatial normalization on encoder features [B, T, D]
x = x - gamma * x.mean(dim=1, keepdim=True)
x = x / (x.std(dim=1, keepdim=True) + 1e-6)
```



(a) *Simpler projection layer for REPA.* Standard MLP projection layer in REPA (middle) loses spatial information while transferring features from target representation (left) to diffusion features. Instead using a simpler convolution layer leads to better spatial information transfer (right).



(b) *Spatial normalization layer.* Patch tokens of pretrained vision encoders have a global component which limits spatial contrast. This causes the tokens in one semantic region (e.g., tomato) to show quite decent cosine similarity with unrelated tokens (e.g., background or cucumber). We hypothesize that we can sacrifice this global information to improve the spatial contrast between patch tokens - leading to better generation performance.



(c) *Overall impact of improved training recipe (iREPA) on spatial structure of diffusion features.*

Figure 6: Motivating two straightforward modifications (iREPA) to enhance spatial feature transfer from target representation to diffusion features. All results reported with SiTL/2 with REPA at 100K.

4.1 EXPERIMENTS

We evaluate the improved training recipe through extensive experiments on Imagenet 256×256 . The experiments address three questions that connect the proposed modifications to convergence, scalability, and compatibility with existing REPA-style recipes:

- Does iREPA improve convergence speed over REPA across diverse external representations? (Figure 7, 8, 12, 13, 15)
- Does iREPA scale across model sizes and alignment depths? (Table 1 [a,b,c], 2, 4, 5, 6, and, Figure 12, 16)
- Does iREPA remain effective for recent representation-alignment recipes such as REPA-E (Leng et al., 2025a), MeanFlow w/ REPA (Geng et al., 2025), and pixel-space diffusion models such as JiT w/ REPA (Li & He, 2025b)? (Table 3 [a,b])

Convergence Speed. We evaluate the convergence behavior of iREPA across diverse vision encoders (DINOv3-B, WebSSL-1B, PE-Core-G, CLIP-L, MoCov3, PE-Lang-G) and model sizes (SiT-XL/2 and SiT-B/2). Fig. 7 shows that iREPA consistently achieves faster convergence than baseline REPA across variations in both target representation and model size.

Target representation. We analyze the behavior of iREPA across different vision encoders in Fig. 8 and Table 4. iREPA improves generation quality across all reported vision encoders. Additional comparisons across all 27 encoders are provided in Appendix C, E.

Encoder size. We analyze iREPA across different encoder sizes. Table 1a reports results for PE-B (90M), PE-L (320M), and PE-G (1.88B). iREPA improves performance across all encoder sizes, and the percentage improvement increases with encoder size (22.2% for PE-B, 38.8% for PE-L, 39.6% for PE-G).

Scalability. We analyze the scalability of iREPA across different model scales in Table 1b. The spatial modifications consistently improve performance, with larger percentage gains for larger models, indicating that the benefit of spatial-information transfer scales with model size in these experiments.

Encoder depth. Table 1c analyzes iREPA across different alignment depths. All results use SiT-B/2 at 100K iterations with DINOv3-B as the target representation. iREPA improves over baseline REPA at each reported alignment depth.

Enc. Size	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Rc. $\uparrow$
PE-B (90M)	59.6	22.5	6.23	0.63	0.60
+iREPA	73.3	17.5	6.04	0.66	0.62
PE-L (320M)	48.6	28.7	6.53	0.59	0.60
+iREPA	76.3	17.6	6.36	0.65	0.61
PE-G (1.88B)	42.7	32.3	6.70	0.57	0.59
+iREPA	70.8	19.5	6.35	0.64	0.61

(a) Vision Encoder size

Model Size	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Rc. $\uparrow$
SiT-B	27.5	49.50	7.00	0.46	0.59
+iREPA	34.1	43.37	7.87	0.50	0.60
SiT-L	55.7	24.10	6.25	0.62	0.60
+iREPA	66.9	20.28	6.14	0.63	0.62
SiT-XL	70.3	19.06	5.83	0.65	0.61
+iREPA	77.9	16.96	6.26	0.66	0.61

(b) Model size

Aln. Depth	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Rc. $\uparrow$
Layer 4	28.1	49.0	6.83	0.46	0.60
+iREPA	41.7	36.0	6.61	0.52	0.62
Layer 6	26.9	50.6	7.00	0.46	0.59
+iREPA	42.4	35.3	6.90	0.53	0.62
Layer 8	24.7	54.8	7.35	0.44	0.58
+iREPA	38.3	38.9	7.19	0.51	0.62

(c) Encoder depth (SiT-B/2)

Table 1: **Variation in training settings.** We find that iREPA leads to consistent gains over baseline REPA across diverse training settings. Unless otherwise specified all results are reported using Dinov2 as encoder, SiTXL/2, 100K steps and vanilla-REPA as baseline.

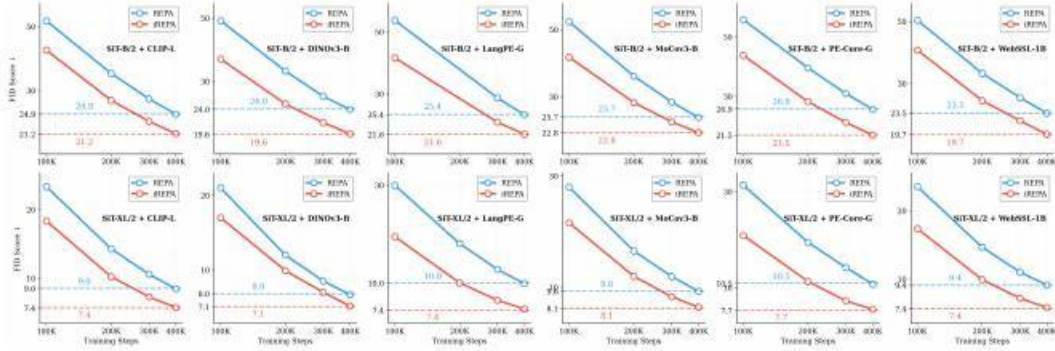


Figure 7: **Accelerating spatial features helps consistently improve convergence speed.**

Ablation on different components. We study the contribution of each iREPA component in Table 2. Both spatial normalization and the convolutional projection improve generation quality over baseline REPA, and the best results are obtained when both components are used together.

Training recipe. Finally, we analyze iREPA across training recipes such as REPA-E (Leng et al., 2025a) and MeanFlow w/ REPA (Geng et al., 2025). Table 3 shows that the spatial modifications in iREPA lead to convergence gains across these recipes.

Classifier-free guidance. We evaluate iREPA with classifier-free guidance (CFG) in Table 4, which reports results across different vision encoders at 400K training iterations. Across encoders, iREPA leads to faster convergence both with and without classifier-free guidance.

Pixel-space diffusion (JiT). We also evaluate iREPA on pixel-space diffusion models such as JiT-B (Li & He, 2025b). Figure 9 compares REPA and iREPA using JiT (Li & He, 2025b). Accelerating spatial-information transfer achieves faster convergence across several vision encoders (e.g., DiNOv2, DiNOv3, PE etc.).

- **Finding 3.** *Accelerating the transfer of spatial structure from target representations to diffusion features improves convergence speed with representation alignment (REPA).*

5 RELATED WORK

We summarize the most relevant related work here and provide a more detailed discussion in Appendix H.

Representation alignment for generation. Many recent works use external representations to improve diffusion model training (Pernias et al., 2023; Fuest et al., 2024). In particular, (Yu et al., 2024; Yao & Wang, 2025; Leng et al., 2025a;b; Kouzelis et al., 2025) show that substantial performance gains can be obtained by aligning internal diffusion features with clean image features from a pretrained vision encoder. (Zhang et al., 2025; Wu et al., 2025) extend this idea to video generation and 3D generation, respectively, and (Ma et al., 2025a) shows that representation alignment can

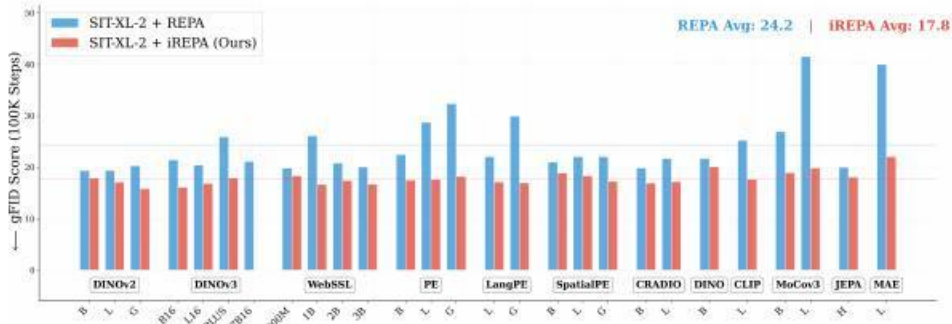


Figure 8: **Variation in target representation.** Across all 27 vision encoders, we find that accentuating transfer of spatial features from target representation to diffusion features (iREPA) leads to consistent improvements in convergence speed. See Appendix C for more results across diverse settings.

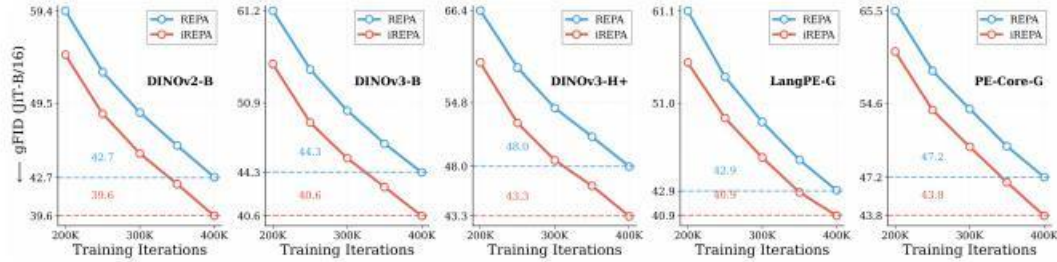


Figure 9: **Convergence comparison with pixel-space diffusion (JiT).** FID convergence curves comparing REPA vs iREPA with pixel space JiT (Li & He, 2025b) across different vision encoders. Accentuating transfer of spatial structure helps consistently improve the convergence speed of across different vision encoders for pixel-space diffusion models. All results are reported with JiTB/16 (Li & He, 2025b), 256 batch size and without classifier-free guidance. Refer Table 10 for further results.

improve the training of unified models. Despite these empirical successes, the mechanisms through which self-supervised features benefit diffusion training remain only partially understood. Our work isolates a specific representational property—spatial structure in patch-token features—and uses this analysis to derive an improved training recipe.

Spatial vs global information tradeoff in pretrained vision encoders. Recent work studies the tradeoff between global and spatial information in pretrained vision encoders (Bolya et al., 2025; Siméoni et al., 2025). (Siméoni et al., 2025) show that continued training of self-supervised vision representations can increase the similarity between the global CLS token and patch tokens, which degrades performance on dense spatial tasks. (Bolya et al., 2025; Heinrich et al., 2024) explicitly train spatially tuned models for dense spatial tasks. We connect this tradeoff to generative training by showing that, for REPA-style alignment, the spatial structure of a vision encoder is more predictive of generation performance than its global discriminative accuracy. This perspective suggests that future representation selection and representation pretraining for generation should account for spatial feature geometry rather than relying only on global semantic benchmarks.

6 CONCLUSION

This paper studies which property of a target representation best explains the effectiveness of representation alignment: global semantic information or spatial structure among patch tokens. Through a large-scale empirical analysis, we show that spatial structure is a substantially stronger predictor of REPA performance than ImageNet-1K linear probing accuracy. We then translate this finding into two simple modifications that accentuate the transfer of spatial information from the target representation to diffusion features. The resulting method, iREPA, consistently improves convergence speed across diverse vision encoders, model scales, and training recipes. We hope

Method	DINOv2-B			DINOv3-B			WebSSL-1B			PE-Core-G		
	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓
Baseline REPA	19.06	70.3	5.83	21.47	63.4	6.19	26.10	53.0	6.90	32.35	42.7	6.70
iREPA (w/o spatial norm)	18.52	73.3	6.11	17.76	74.7	5.81	21.17	64.6	6.27	24.97	57.4	6.21
iREPA (w/o conv proj)	17.66	72.8	6.03	18.28	70.8	6.18	18.44	71.0	6.22	21.72	61.5	6.26
iREPA (full)	16.96	77.9	6.26	16.26	78.8	6.14	16.66	77.5	6.18	18.19	75.0	6.03

Table 2: **Ablation on different components.** We see that across different encoders - both spatial normalization layer and convolution projection layer (§4) lead to significant gains in convergence speed; with best results obtained using both. All results reported at 100K steps and SiT-XL/2.

Encoder	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑	w/o CFG				w/ CFG			
						4 NFE		1 NFE		4 NFE		1 NFE	
						IS↑	FID↓	IS↑	FID↓	IS↑	FID↓	IS↑	FID↓
WebSSL-1B	52.8	26.5	5.20	0.620	0.585								
+iREPA-E	87.0	13.2	5.28	0.699	0.598								
PE-G	50.9	25.9	5.68	0.612	0.576								
+iREPA-E	80.0	16.4	5.40	0.667	0.616								
DINOv3-B	82.2	14.4	4.68	0.694	0.596								
+iREPA-E	93.6	11.7	4.57	0.703	0.613								
DINOv2-B	87.5	12.9	5.40	0.708	0.586								
+iREPA-E	91.3	12.1	4.86	0.712	0.602								

Encoder	w/o CFG				w/ CFG			
	4 NFE		1 NFE		4 NFE		1 NFE	
	IS↑	FID↓	IS↑	FID↓	IS↑	FID↓	IS↑	FID↓
WebSSL-1B	27.22	51.40	24.14	58.67	87.85	16.59	69.06	23.74
+iREPA	31.48	45.67	27.33	55.70	100.74	13.89	78.67	20.69
DINOv3-B	28.36	49.58	25.47	57.01	93.30	15.56	72.41	22.55
+iREPA	33.63	44.52	29.67	53.75	124.54	11.05	98.93	17.32
DINOv2-B	28.77	48.87	25.43	56.63	94.40	15.28	71.95	22.17
+iREPA	33.63	44.52	29.67	53.75	111.44	12.59	90.25	18.62

(a) REPA-E (SiT-XL/2)

(b) MeanFlow w/ REPA (SiT-B/2)

Table 3: **Variation in training recipes.** We apply our spatial improvements on top of REPA-E (a) and Meanflow with REPA (b); achieving consistent gains. For Meanflow, we report results at 1 and 4 NFEs. CFG value of 2.0 is used for Meanflow. All results are reported at 100K training iterations.

these results motivate future work to study representation alignment through the geometry of spatial features and to design target representations that are selected and trained specifically for generative modeling.

Vision Encoder	Steps	w/o CFG					w/ CFG				
		IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
DINOv2-B	100K	69.20	19.3	5.89	0.64	0.61	157.2	6.35	5.91	0.769	0.536
+IREPA	100K	77.92	16.9	6.26	0.66	0.61	179.3	5.15	6.23	0.783	0.544
DINOv2-B	400K	127.4	7.76	5.06	0.70	0.66	263.0	1.98	4.60	0.799	0.610
+IREPA	400K	128.6	7.52	4.89	0.71	0.65	268.8	1.93	4.59	0.799	0.600
DINOv3-B	100K	63.64	21.4	6.14	0.63	0.60	144.0	7.57	6.09	0.762	0.526
+IREPA	100K	78.79	16.2	6.14	0.66	0.61	181.9	4.87	6.10	0.780	0.547
DINOv3-B	400K	126.7	8.10	5.06	0.70	0.66	261.2	1.99	4.58	0.799	0.609
+IREPA	400K	132.9	7.13	4.93	0.71	0.66	272.4	1.89	4.58	0.799	0.600
WebSSL-1B	100K	53.87	25.5	6.57	0.61	0.59	124.0	9.59	6.37	0.756	0.515
+IREPA	100K	77.47	16.6	6.18	0.66	0.61	177.6	5.09	6.11	0.787	0.538
WebSSL-1B	400K	116.9	9.39	5.14	0.70	0.64	250.5	2.24	4.61	0.809	0.580
+IREPA	400K	130.8	7.48	4.91	0.70	0.65	271.6	1.90	4.58	0.798	0.609
PE-Core-G	100K	42.74	32.3	6.70	0.57	0.59	97.2	14.1	6.56	0.714	0.525
+IREPA	100K	75.01	18.1	6.03	0.64	0.61	176.8	5.66	6.08	0.771	0.544
PE-Core-G	400K	109.4	10.4	5.00	0.69	0.64	238.4	2.44	4.57	0.805	0.585
+IREPA	400K	132.0	7.78	5.02	0.70	0.65	275.4	1.93	4.59	0.796	0.606

Table 4: **Results across different encoders with and w/o classifier-free guidance.** SiT-XL/2 is used base model. See Appendix E for detailed results on SiT-B/2 & SiT-L/2 across all encoders.

REPRODUCIBILITY STATEMENT

We provide all implementation details and hyperparameters in Appendix G. Our code, model checkpoints, and analysis results will be released upon acceptance.

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A DETAILED CORRELATION ANALYSIS

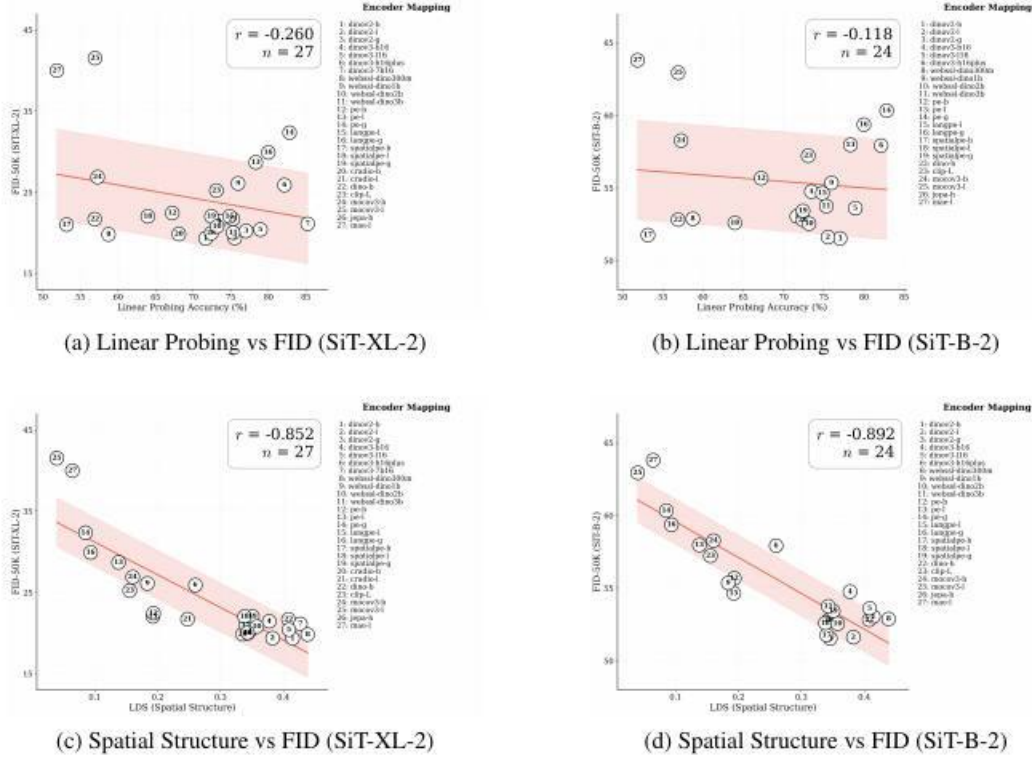


Figure 10: Detailed correlation analysis with encoder labels. The top row reports the correlation between linear probing accuracy and FID for SiT-XL-2 and SiT-B-2. The bottom row reports the corresponding correlation between spatial structure (LDS) and FID. Each point is labeled with its encoder index (see legend). Across both model scales, the spatial-structure metric correlates substantially more strongly with generation quality than linear probing accuracy.

B SPATIAL SELF-SIMILARITY METRICS



Figure 11: **Explaining semantic region self-similarity (SRSS) metric.** Visual explanation of SRSS metric: For each image, we compute a mask using SAM2 and select anchor-positive-negative triplets. The SSM measures the difference in cosine similarity between anchor-positive pairs (within mask) and anchor-negative pairs (outside mask). Intuitively, larger values indicate that patches on the same semantic region are more similar than patches on unrelated regions — indicating better spatial structure preservation.

Setup. Given an image $\mathcal{I}$ and vision encoder $\mathcal{E}$, we extract patch tokens $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ with $T = H \times W$ and indices $t \in [T]$ placed on an $H \times W$ lattice. Let $d : [T] \times [T] \rightarrow \mathbb{N}$ be the Manhattan distance, and let the (cosine) self-similarity kernel be

$$K_{\mathcal{X}}(t, t') = \frac{\langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2} \in [-1, 1],$$

a standard local self-similarity measure (Shechtman & Irani, 2007).

- *Patch token representation:* $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ where $T = H \times W$ is the spatial grid of patches (patch index set $[T]$). Let $d : [T] \times [T] \rightarrow \mathbb{N}$ denote Manhattan distance on the $H \times W$ grid.
- *Self-similarity kernel:* $K : [T] \times [T] \rightarrow \mathbb{R}$ measures self-similarity (Shechtman & Irani, 2007) between patch tokens. We use the cosine kernel $K_{\mathcal{X}}(t, t') = \langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle / (\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2)$.
- *Spatial self-similarity metric:* a functional $m : \mathcal{K} \times \mathcal{D} \rightarrow \mathbb{R}$ measures how self-similarity K between patch tokens varies with lattice distance d . Larger values indicate stronger spatial organization, with nearby patches more similar to one another than distant patches.

We next define several straightforward metrics for measuring the spatial self-similarity structure of the target representations.

Local vs. Distant Similarity (LDS). We first consider a simple correlogram contrast (local vs. distant) metric (Huang et al., 1997):

$$\text{LDS}(\mathcal{X}) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') < r_{\text{near}}] - \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') \geq r_{\text{far}}],$$

where r_{near} and r_{far} are hyperparameters. By default, we use $r_{\text{near}} = r_{\text{far}} = H/2$, and the correlations are robust to their exact choices. Larger values indicate that, *on average*, nearby patches are more similar than distant patches.

Correlation Decay Slope (CDS). Given the patch tokens $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$, we compute the spatial correlogram $g_{\mathcal{X}}(\delta) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') = \delta]$ for distances $\delta \in \{1, \dots, \Delta\}$. We then fit a least-squares line $\hat{g}_{\mathcal{X}}(\delta) \approx \alpha + \beta\delta$ and define:

$$\text{CDS}(\mathcal{X}) = -\hat{\beta}$$

where $\hat{\beta}$ is the fitted slope. Larger CDS values indicate faster similarity decay with distance and therefore stronger spatial organization.

Semantic-Region Self-Similarity (SRSS). For each image, we first sample a binary mask $\mathbf{M} \in \mathbb{R}^{H \times W}$ containing a random object using SAM2 (Ravi et al., 2024) and downsample it to $H \times W$. We then select anchor, positive, and negative points. The anchor and positive points are sampled from within the mask, where they should have related semantics, while the negative point is sampled from outside the mask, where it should be less related to the anchor. If the encoder representation preserves spatial structure, the anchor-positive pair should have higher cosine similarity than the anchor-negative pair. We therefore define the Spatial Structure Metric (SSM) as:

$$\text{SSM}(\mathbf{P}) = \mathbb{E}_{\text{anchor} \in \mathbf{M}} [\cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{pos}}) - \cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{neg}})]$$

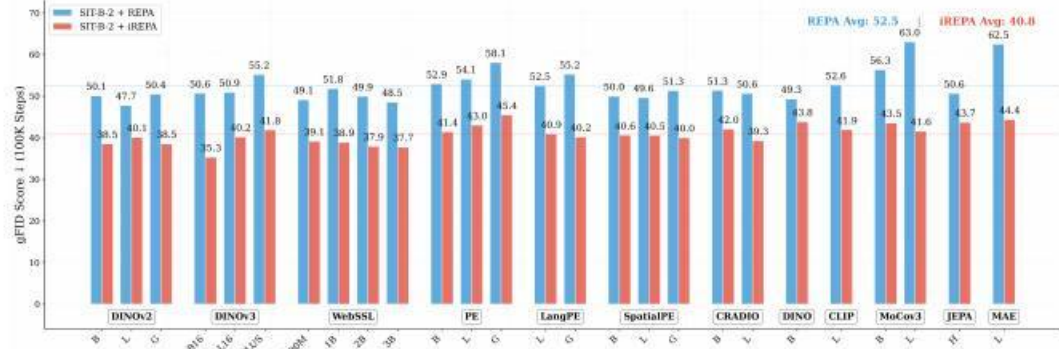
where positive points are within Manhattan distance $d \leq r_{near}$ from the anchor within the mask, and negative points are at distance $d \geq r_{far}$ outside the mask.

RMS Spatial Contrast (RMSC). Finally, we consider a simple contrast metric to capture the spatial contrast between patch token representations. Given normalized features $\hat{\mathbf{x}}_t = \mathbf{x}_t / \|\mathbf{x}_t\|_2$ for each patch t , we compute:

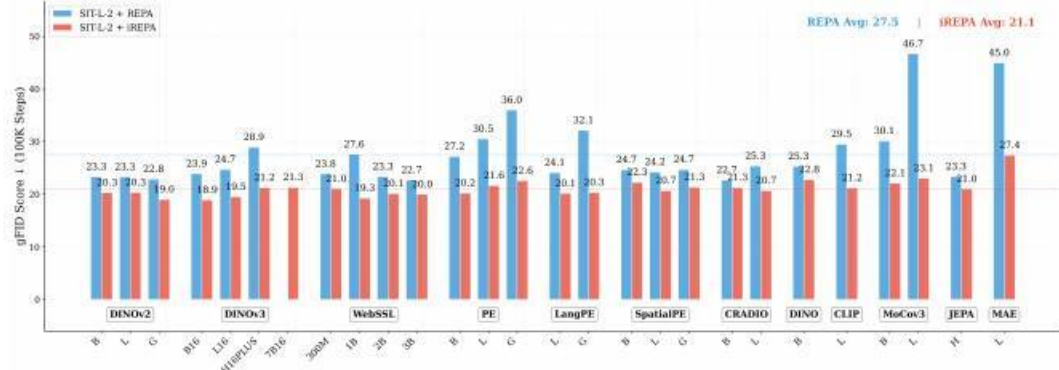
$$\text{RMSC}(\mathcal{X}) = \sqrt{\frac{1}{T} \sum_{t=1}^T \|\hat{\mathbf{x}}_t - \bar{\mathbf{x}}\|_2^2}$$

where $\bar{\mathbf{x}} = \frac{1}{T} \sum_{t=1}^T \hat{\mathbf{x}}_t$ is the mean of normalized features across all spatial locations. Higher RMSC values indicate greater spatial diversity in the feature representations and therefore stronger spatial structure, whereas lower values indicate more uniform feature distributions consistent with reduced spatial contrast, as can occur when global information diminishes spatial structure (Siméoni et al., 2025).

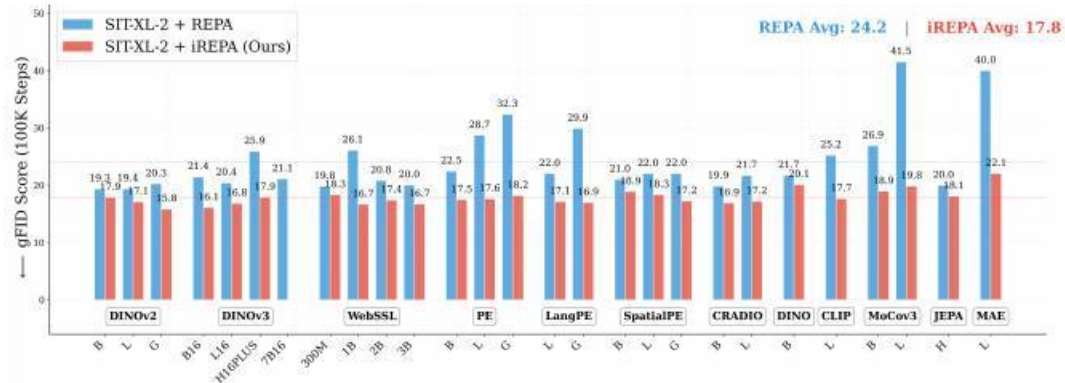
C VISION ENCODER VARIATION



(a) SiT-B-2 (130M parameters)



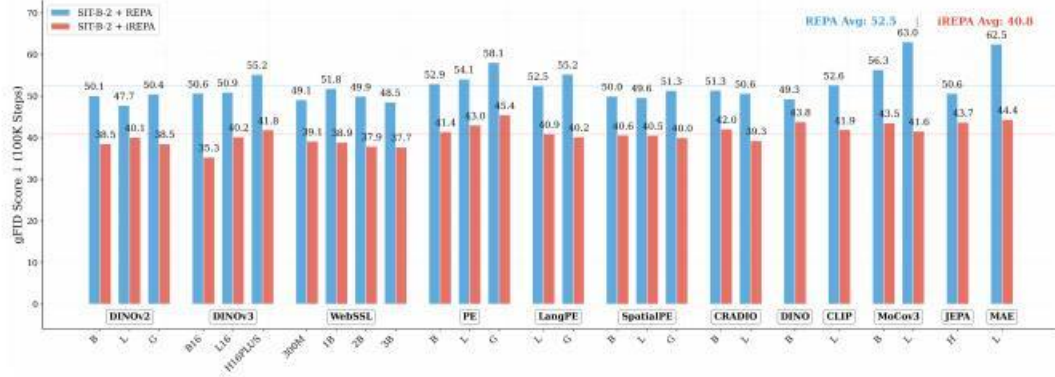
(b) SiT-L-2 (458M parameters)



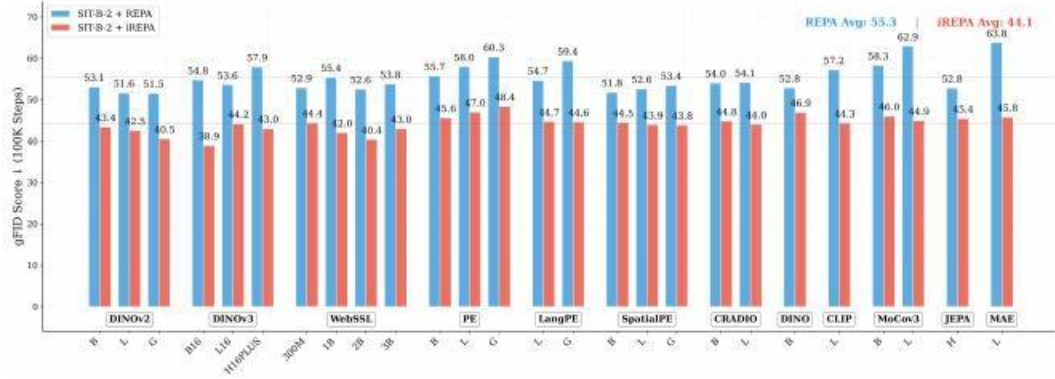
(c) SiT-XL-2 (675M parameters)

Figure 12: **Variation in vision encoders.** Ablation studies showing iREPA improvements across different vision encoders for SiT-B-2, SiT-L-2, and SiT-XL-2 models. iREPA consistently improves generation quality across all encoders and model sizes.

D ENCODER DEPTH VARIATION



(a) SiT-B-2 with 6 encoder layers



(b) SiT-B-2 with 8 encoder layers

Figure 13: **Effect of encoder depth on iREPA performance.** Comparison of SiT-B-2 performance with different encoder depths (6 vs 8 layers) across various vision encoders. The results show consistent improvements with iREPA regardless of encoder depth, with slightly better performance observed with 8 encoder layers. Both configurations demonstrate significant FID improvements when spatial regularization is applied.

E COMPREHENSIVE RESULTS ACROSS VISION ENCODERS

E.1 SiT-B/2 RESULTS

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
CLIP-ViT-L	59.44	24.84	6.46	0.584	0.645
+iREPA	69.70	21.26	6.64	0.601	0.644
DINOv2-B	64.92	22.75	6.54	0.593	0.653
+iREPA	70.88	21.40	6.77	0.597	0.653
DINOv2-L	66.65	22.46	6.53	0.594	0.650
+iREPA	72.41	20.99	6.84	0.600	0.650
DINOv3-B16	62.57	23.91	6.77	0.584	0.647
+iREPA	75.48	19.65	6.68	0.606	0.649
DINOv3-H16+	61.91	24.82	6.57	0.579	0.645
+iREPA	71.76	21.80	6.84	0.592	0.655
DINO-B	58.37	24.71	6.32	0.586	0.635
+iREPA	57.55	24.41	6.43	0.587	0.638
LangPE-G	58.42	25.37	6.32	0.583	0.641
+iREPA	66.88	21.62	6.65	0.604	0.649
MoCov3-B	57.61	25.60	6.25	0.579	0.640
+iREPA	61.41	22.80	6.46	0.596	0.639
PE-B	58.86	25.78	6.44	0.58	0.64
+iREPA	69.50	21.15	6.66	0.601	0.659
PE-G	56.09	26.84	6.45	0.568	0.641
+iREPA	70.43	21.52	6.58	0.598	0.651
SpatialPE-B	59.19	24.28	6.37	0.586	0.646
+iREPA	61.50	22.71	6.41	0.604	0.644
SpatialPE-L	62.13	23.87	6.51	0.59	0.65
+iREPA	68.83	21.07	6.59	0.607	0.651
WebSSL-1B	63.20	23.48	6.33	0.593	0.646
+iREPA	73.54	19.74	6.63	0.612	0.646
WebSSL-2B	66.56	22.27	6.69	0.596	0.654
+iREPA	72.13	20.47	6.89	0.609	0.648
WebSSL-300M	62.06	23.64	6.48	0.590	0.646
+iREPA	66.83	21.46	6.71	0.606	0.653
WebSSL-3B	65.91	22.55	6.53	0.598	0.654
+iREPA	71.98	20.64	6.71	0.607	0.650

Table 5: **SiT-B/2 performance across vision encoders at 400K iterations.** iREPA consistently improves generation quality. All baselines are reported using vanilla-REPA (Yu et al., 2024) for training.

E.2 SiT-L/2 RESULTS

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
CLIP-ViT-L	107.7	10.6	5.20	0.682	0.643
+iREPA	115.8	9.41	5.15	0.689	0.653
DINOv2-B	113.3	9.61	5.11	0.684	0.653
+iREPA	113.9	9.36	5.11	0.690	0.655
DINOv3-B16	115.3	9.41	5.24	0.687	0.651
+iREPA	118.1	9.06	5.25	0.685	0.650
DINOv3-H16+	110.7	10.6	5.32	0.677	0.652
+iREPA	120.5	9.64	5.29	0.681	0.663
LangPE-G	102.8	11.2	5.13	0.684	0.637
+iREPA	112.7	9.37	5.02	0.690	0.653
MoCov3-B	102.1	11.0	5.14	0.685	0.635
+iREPA	104.7	10.1	5.00	0.690	0.641
PE-G	98.38	12.6	5.31	0.670	0.646
+iREPA	117.1	9.65	5.33	0.683	0.651
WebSSL-1B	107.9	10.5	5.13	0.686	0.644
+iREPA	121.5	8.62	5.01	0.696	0.653

Table 6: **SiT-L/2 performance across vision encoders at 400K iterations.** iREPA consistently improves generation quality across all encoders, with particularly strong gains in FID and IS metrics. All baselines are reported using vanilla-REPA (Yu et al., 2024) for training.

F ADDITIONAL EXPERIMENTAL RESULTS

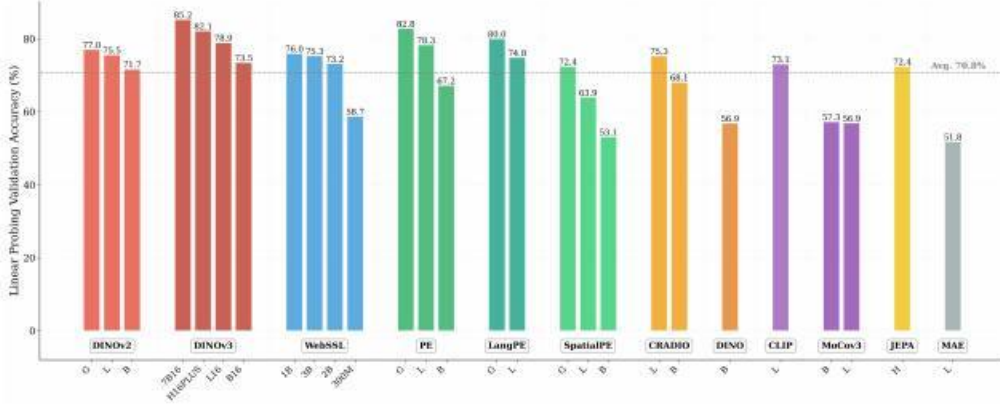


Figure 14: **Global information in mean patch tokens.** We find that in addition to [CLS] token, mean of patch tokens also contains substantial global semantic information. While this helps improve global performance, it reduces the contrast between individual patch tokens, potentially hindering spatial structure transfer. We find that we can remove some of this global information (through mean of patch tokens) to improve spatial structure transfer during representation alignment (§4).

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
DINOv2-B	262.97	1.98	4.60	0.799	0.610
+iREPA (Ours)	268.79	1.93	4.59	0.799	0.600
DINOv3-B	261.18	1.99	4.58	0.799	0.609
+iREPA (Ours)	272.41	1.89	4.58	0.799	0.600
WebSSL-1B	250.53	2.24	4.61	0.809	0.580
+iREPA (Ours)	271.59	1.90	4.58	0.798	0.609
PE-G	238.37	2.44	4.57	0.805	0.585
+iREPA (Ours)	275.36	1.93	4.59	0.796	0.606

Table 7: **Generation quality with classifier-free guidance.** Comparison of REPA vs iREPA with CFG (scale 2.0) across different vision encoders. All experiments use SiT-XL/2 trained for 400K iterations with 250 sampling steps. iREPA consistently improves both IS and FID metrics across all encoders.

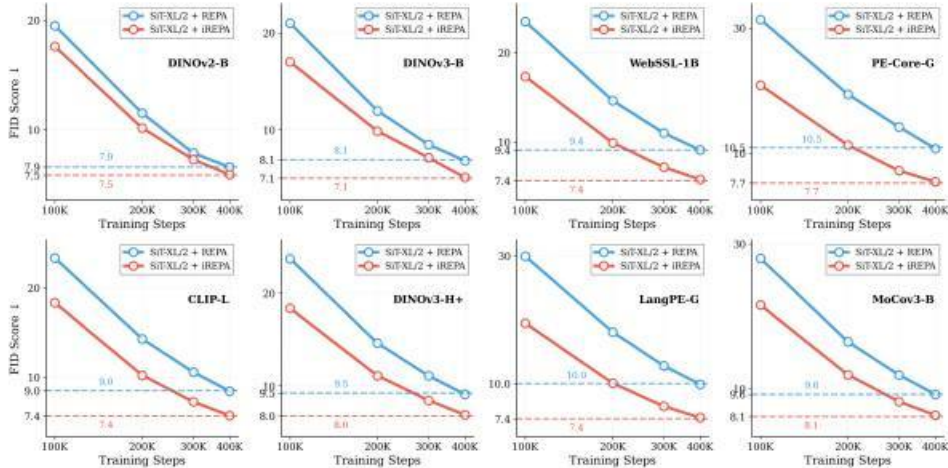


Figure 15: **Accentuating spatial features helps consistently improve convergence speed.** Results for SiT-XL/2 with REPA and iREPA.

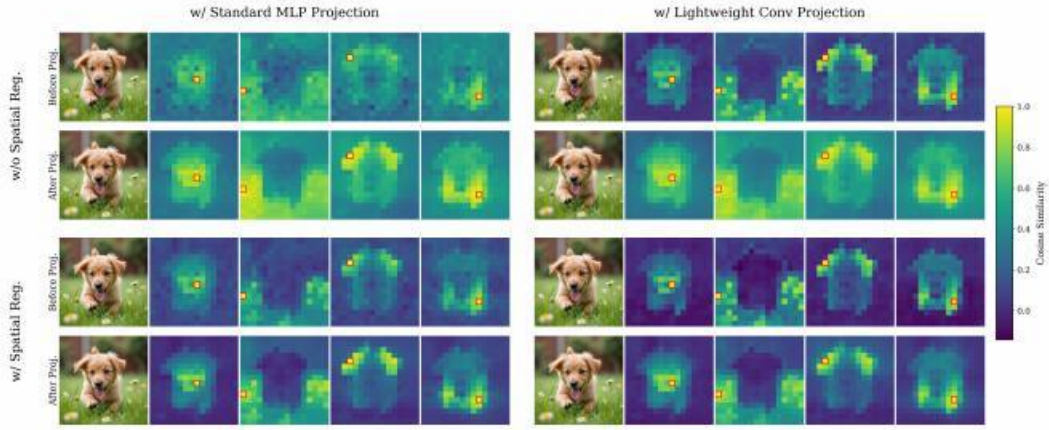


Figure 16: **Visualizing the impact of two straightforward improvements to enhance spatial feature transfer.** **First**, we find that standard MLP projection layer (top-left) losses spatial information while transferring features from the pretrained encoder (after projection) to diffusion model (before projection). Instead using a simpler convolution layer better preserves the spatial information transfer (top-right). **Second**, we observe that vision encoder features often have limited spatial contrast (§2. This causes the tokens in one semantic region (*e.g.*, dog) to show quite decent cosine similarity with unrelated tokens (*e.g.*, background). We address this with a simple spatial regularization layer, which accentuates the spatial contrast in the learned representation — leading to better generation performance. Best results are obtained while using both (refer Table 2).

G IMPLEMENTATION DETAILS

Training setup. We follow the same setup as in REPA (Yu et al., 2024) unless otherwise specified. All training is conducted on the ImageNet (Deng et al., 2009) training split. For preprocessing, we adopt the protocol from ADM (Dhariwal & Nichol, 2021), where each image is center-cropped and resized to 256×256 resolution. We use `stabilityai/sd-vae-ft-mse` VAE (AI, n.d.) throughout our diffusion training and inference. For spatial normalization layer we use $\gamma \in [0.6, 0.8]$ and for projection layer we use a convolutional layer with kernel size 3 and padding 1. For optimization, we use AdamW (Kingma & Ba, 2014; Loshchilov, 2017) with a constant learning rate of 1×10^{-4} , and a global batch size of 256. During training, we use `bfloat16` mixed precision and `torch.compile` to accelerate training, and gradient clipping and exponential moving average (EMA) to the generative model for stable optimization.

For REPA-E (Leng et al., 2025a) and JiT (Li & He, 2025a) experiments, we use the official open-source implementation. For MeanFlow (Geng et al., 2025) experiments, we received the implementation (including REPA) from original authors and adapt it to introduce the two straightforward changes for iREPA (§4).

Evaluation. For image generation evaluation, we strictly follow the ADM setup (Dhariwal & Nichol, 2021). We report generation quality using Fréchet inception distance (gFID) (Heusel et al., 2017), structural FID (sFID) (Nash et al., 2021), inception score (IS) (Salimans et al., 2016), precision (Prec.) and recall (Rec.) (Kynkäänniemi et al., 2019), measured on 50K generated images. For sampling, we follow the approach in REPA (Yu et al., 2024), using the SDE Euler-Maruyama sampler with 250 steps. For JiT (Li & He, 2025a), we use 50 inference steps following official implementation.

H MORE DISCUSSION ON RELATED WORK

Classical spatial features in computer vision. Spatial feature extraction has long been fundamental to computer vision. Classical methods like SIFT (Lowe, 1999), HOG (Dalal & Triggs, 2005), SURF (Bay et al., 2006), and ORB (Rublee et al., 2011) providing robust local descriptors for tasks ranging from object detection to image matching. While these handcrafted features excel at capturing local spatial patterns and geometric invariances, recent work in generative modeling has primarily focused on representations from modern self-supervised methods that demonstrate strong global classification performance, such as DINOv2 (Oquab et al., 2024) and CLIP (Radford et al., 2021). Our findings suggest a different perspective: since spatial structure preservation is critical for generation quality, even classical spatial features could potentially improve diffusion training when properly aligned. This highlights the potential of leveraging the full spectrum of spatial feature extractors, from traditional handcrafted descriptors to modern learned representations, provided that they maintain strong spatial coherence.

Pretrained visual encoders for generative models. Pretrained visual encoders have supported generative models in several capacities: as discriminators to accelerate GAN convergence (Goodfellow et al., 2020; Sauer et al., 2021; 2022; 2023; Radford et al., 2021), as teachers in adversarial distillation for diffusion models (Sauer et al., 2024; Kang et al., 2024), and more recently as alignment targets. In GANs (Goodfellow et al., 2020), pretrained features have not only improved convergence speed but also enabled scaling to large datasets, as demonstrated by StyleGAN-XL (Sauer et al., 2022) and StyleGAN-T (Sauer et al., 2023) with CLIP (Radford et al., 2021) features. For diffusion models, adversarial distillation leverages pretrained encoders to guide student networks toward higher-fidelity samples, showing clear improvements in perceptual quality. In particular, REPA (Yu et al., 2024) aligns diffusion features with external encoders, demonstrating that representation alignment can improve both generation convergence and quality. Building on this direction, we focus not only on alignment but specifically on spatial structure perseverance rather than their discriminative capabilities.

Denoising transformers. Transformer architectures have become the dominant backbone for scalable generative modeling, with various formulations including diffusion transformers (Peebles & Xie, 2023) and flow matching variants (Ma et al., 2024). Recent architectures like GenTron (Chen et al., 2023) scale transformers to over 3B parameters for text-to-image synthesis. U-ViT (Bao et al., 2023) demonstrates that pure transformer backbones without U-Net inductive biases can achieve competitive performance. ARDiT (Tang et al., 2024) and DART (Jing et al., 2024) explore

autoregressive formulations that combine denoising with sequential generation, enabling flexible trade-offs between quality and speed. Despite these architectural advances, training these models from scratch remains computationally expensive, often requiring millions of iterations to achieve good generation quality. While representation alignment methods like REPA (Yu et al., 2024) have shown that pretrained features can dramatically accelerate convergence, the mechanisms behind this improvement remained unclear. Our analysis reveals that the key benefit comes from preserving spatial structure rather than semantic alignment, explaining why certain encoders provide stronger acceleration than others and guiding the design of more effective alignment strategies. While (Wang et al., 2025c) propose improvements to REPA training through early stopping, they hypothesize that “REPA predominantly distills global semantic information while leaving structural information untapped.” In contrast, our analysis reveals that spatial structure (not global semantic information) already plays a very significant role in the effectiveness of REPA (§2, §3). Thus, while we indeed find that spatial structure remains underexploited (§4) in REPA, we surprisingly find that majority of improvements in REPA are already coming from introducing an inductive bias for spatial structure (not global information).

Denoising as self-supervised learning task. The connection between denoising and representation learning has been explored from multiple perspectives. Early work by Abstreiter et al. (2021) extended diffusion objectives for representation learning, demonstrating that denoising naturally learns meaningful features. SODA (Hudson et al., 2024) introduces a diffusion model with an information bottleneck to learn compact representations. Chen et al. (2024) deconstruct diffusion models and find that a simple denoising autoencoder suffices for strong self-supervised performance. Wang & He (2025) introduce a dispersive loss to encourage internal representation diversity in diffusion models, improving generative performance without external encoders. Similarly, Jiang et al. (2025) propose Self-Representation Alignment (SRA) to align a diffusion transformer’s latent features across noise levels, providing self-guidance without an auxiliary model. These works establish denoising as a fundamental self-supervised task that naturally encourages learning of robust features. Our findings complement this view by showing that when diffusion models are aligned with strong spatial representations from self-supervised encoders, both the generative and discriminative capabilities improve, suggesting that spatial structure preservation is a key factor in this synergy.

Scaling self-supervised vision encoders. Recent years have seen remarkable progress in scaling self-supervised vision models to unprecedented sizes and datasets. DINOv3 (Siméoni et al., 2025) trains a 7B parameter ViT on 1.7 billion images without labels by aligning representations from different augmentations. WebSSL (Fan et al., 2025) demonstrates that visual models trained on more than 2 billion images can match language-supervised models like CLIP (Radford et al., 2021) on vision-language tasks without language supervision. C-RADIO (Heinrich et al., 2024) combines multiple teacher models through distillation to create versatile encoders that excel across diverse visual domains. I-JEPA (Assran et al., 2023) explores predictive architectures that learn by predicting masked regions in representation space, instead of reconstructing pixels directly. SAM (Ravi et al., 2024) specializes in promptable segmentation through large-scale supervised training. The Perception Encoder family (Bolya et al., 2025; Cho et al., 2025) shows that intermediate features often outperform final representations for dense prediction tasks. While these models are typically evaluated on global tasks like image classification, our work reveals a crucial insight: strong performance on discriminative benchmarks does not necessarily translate to better generation quality. Instead, we find that encoders preserving spatial structure, regardless of their classification accuracy—provide the most benefit for diffusion training. This suggests that the evaluation metrics for self-supervised encoders should be reconsidered when targeting generative applications, with spatial coherence being as important as semantic understanding.

I NOTE ON LLM USAGE

We use GPT-5 (OpenAI, 2025) for considering different variations of the spatial structure metrics discussed in the paper §3. Furthermore, all figures in the paper are directly generated from our experiment logs and checkpoints using Claude-Code (Anthropic, 2025). Additionally we use LLM help for searching and formulating relevant work in Appendix H. We use cursor in some parts to help with paper writing.

J ADDITIONAL DISCUSSION ON SPATIAL STRUCTURE METRICS

Spatial structure metrics can be used to measure the representation gap. Yu et al. (2024) use linear probing to measure the representation gap, using the increase in validation accuracy to explain the effectiveness of REPA. We find that spatial structure metrics (SSM) can also be used to measure the representation gap and explain the improvements across models. As shown in Figure 17, we see that representation alignment helps close the gap between SSM performance of a pretrained encoder like DINOv2 and the diffusion features.

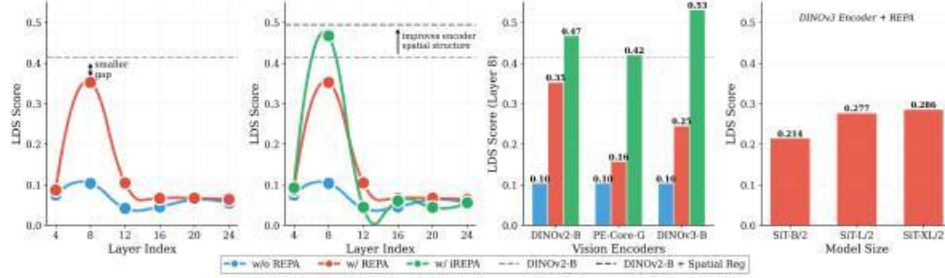


Figure 17: Representation alignment as bridging the spatial feature gap. a) We find that representation alignment can also be seen as bridging the spatial feature gap between diffusion and vision encoder patch features. b) iREPA (§4) accentuates the spatial features of the vision encoder (at cost of some global information) — helping achieve better generation performance. c) iREPA helps consistently improve performance across different vision encoders. d) Spatial structure improvements scale with model size. All results are reported with SiT-XL/2 at 400k iterations.

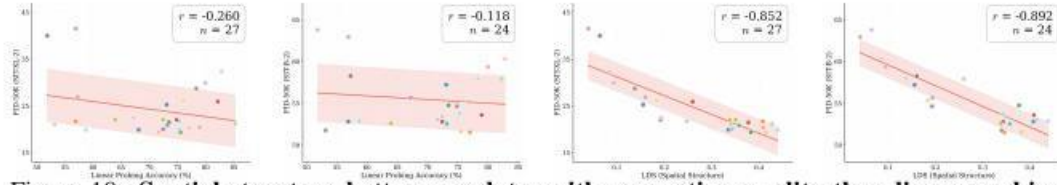


Figure 18: Spatial structure better correlates with generation quality than linear probing. Correlation analysis across 27 vision encoders. **Left two:** Linear probing accuracy vs FID for SiT-XL-2 (Pearson $r = -0.26$) and SiT-B-2 ($r = -0.12$) shows weak correlation. **Right two:** Spatial structure (LDS) vs FID for SiT-XL-2 ($r = -0.85$) and SiT-B-2 ($r = -0.89$) shows strong correlation. See Figure 10 in Appendix for detailed plots with encoder labels.

K ADDITIONAL ANECDOTAL COMPARISONS

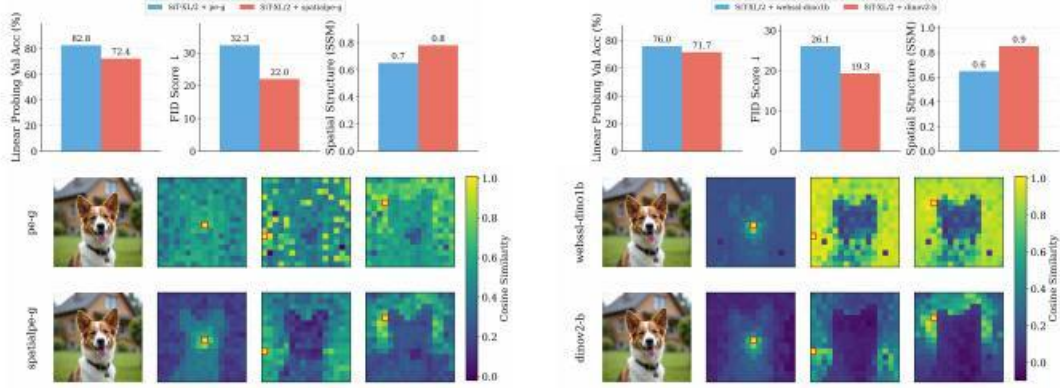


Figure 19: **Motivating anecdotes from recent SSL encoders — spatial structure matters.** **Top row:** Metrics comparison showing inverse relationship between ImageNet accuracy and generation quality. **Left:** PE-g achieves higher accuracy (82.8% vs 72.4%) but worse FID (32.3 vs 22.0) than Spatial-PE-g, with much lower spatial structure (LDS: 0.1 vs 0.4). **Right:** WebSSL-dino1b shows higher accuracy (76.0% vs 71.7%) but worse FID (26.1 vs 19.3) than DINOv2-b, with weaker spatial structure (LDS: 0.2 vs 0.4). **Bottom row:** Spatial cosine similarity visualizations confirm that encoders with better generation (Spatial-PE-g, DINOv2-b) maintain clear spatial coherence, while those optimized for classification (PE-g, WebSSL) lose spatial structure.

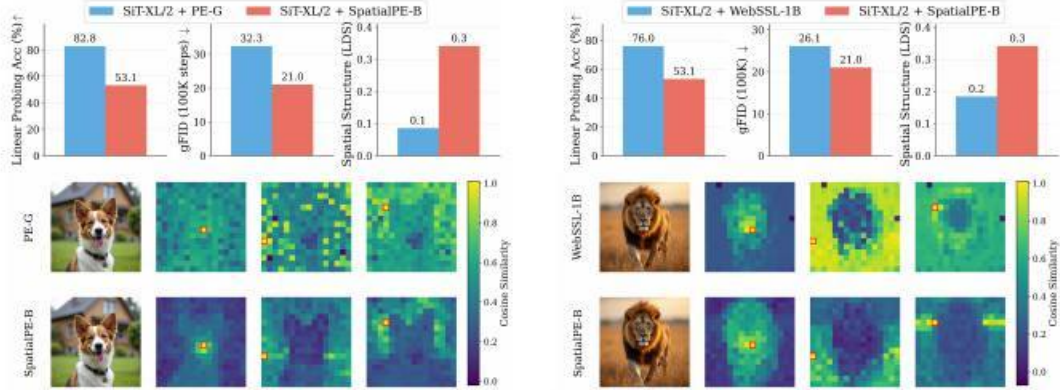


Figure 20: **Motivating examples — spatial structure matters.** **Top:** Metrics comparison showing inverse relationship between ImageNet accuracy and generation quality. **Left:** PE_{core}-G, despite having significantly higher validation accuracy (82.8% vs. 53.1%), shows worse generation quality compared to PE_{Spatial}-B (Bolya et al., 2025). **Right:** Similarly, WebSSL-1B (Fan et al., 2025) also shows much better global performance (76.0% vs. 53.1%), but worse generation. **Bottom:** We find that spatial structure instead provides a better predictor of generation quality than global performance. See §3 for spatial structure metric. All results reported at 100K using SiT-XL/2 and REPA.

L FURTHER DISCUSSION

Role of Spatial Normalization We visualize the impact of spatial normalization on vision encoder representations. Vision encoders can have significant global components or overlays that limit the contrast between spatial tokens. For example, tokens in one semantic region can be highly correlated with tokens in unrelated regions (e.g., background), reducing the spatial distinctiveness of features. Spatial normalization removes this global overlay to enhance contrast between different spatial tokens, allowing the model to better preserve local spatial structure while reducing the dominance of global information that can interfere with generation quality.

The following figures demonstrate this effect across different examples, showing how spatial normalization transforms the feature representations to emphasize spatial structure:

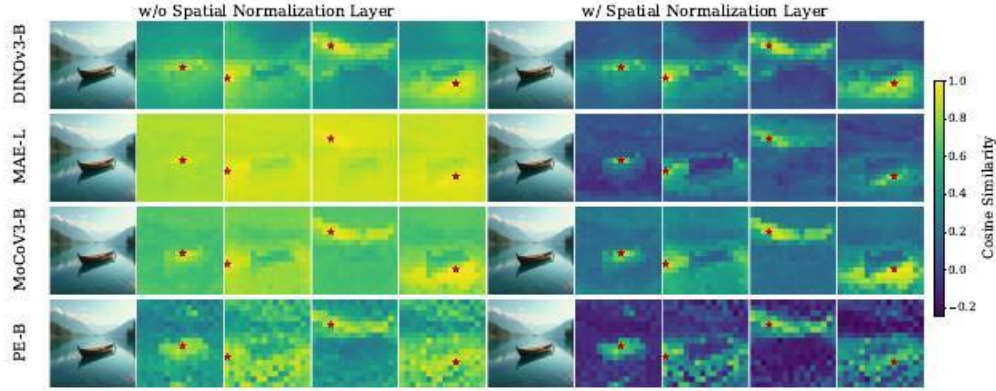


Figure 21: Visualizing impact of spatial normalization (Example 1). The heatmaps show token similarity patterns before and after spatial normalization. Without normalization (left), global components create high correlations across unrelated regions. With spatial normalization (right), local spatial structure is enhanced while reducing global interference, resulting in more distinct semantic boundaries.

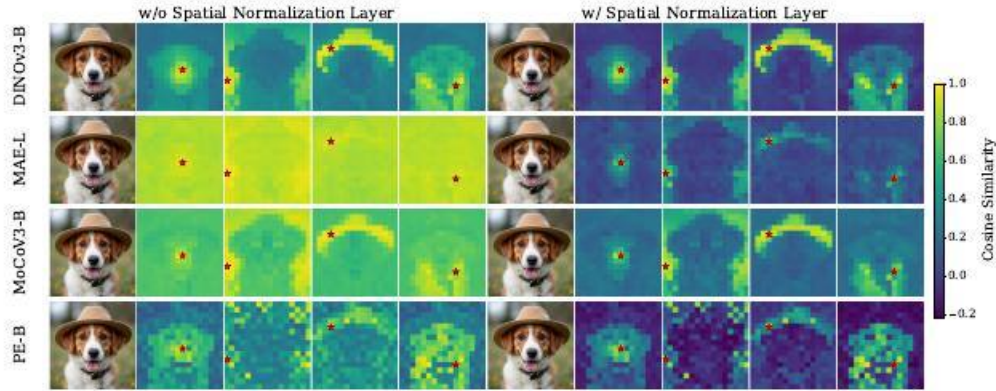


Figure 22: Visualizing impact of spatial normalization (Example 3). The heatmaps show token similarity patterns before and after spatial normalization. Without normalization (left), global components create high correlations across unrelated regions. With spatial normalization (right), local spatial structure is enhanced while reducing global interference, resulting in more distinct semantic boundaries.

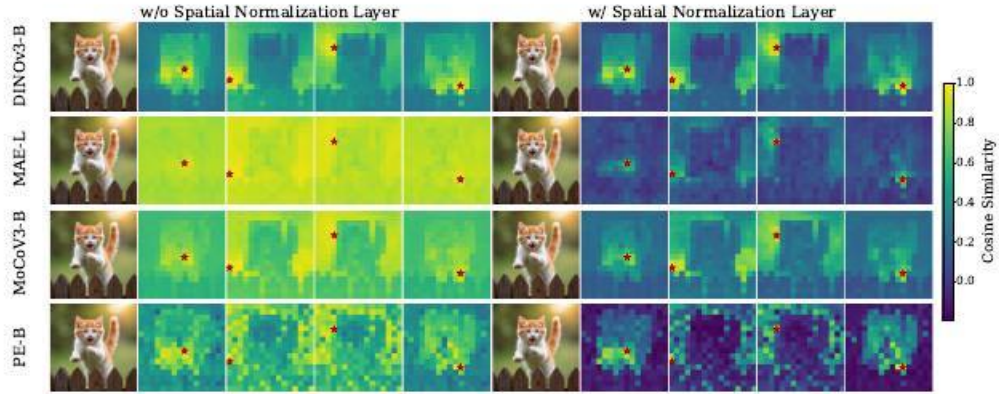


Figure 23: Visualizing impact of spatial normalization (Example 2). Feature similarity maps demonstrate how spatial normalization improves spatial contrast. The original features (left) exhibit a global overlay that reduces distinction between foreground and background regions. After normalization (right), spatial tokens become more locally coherent, with clearer separation between different semantic regions.

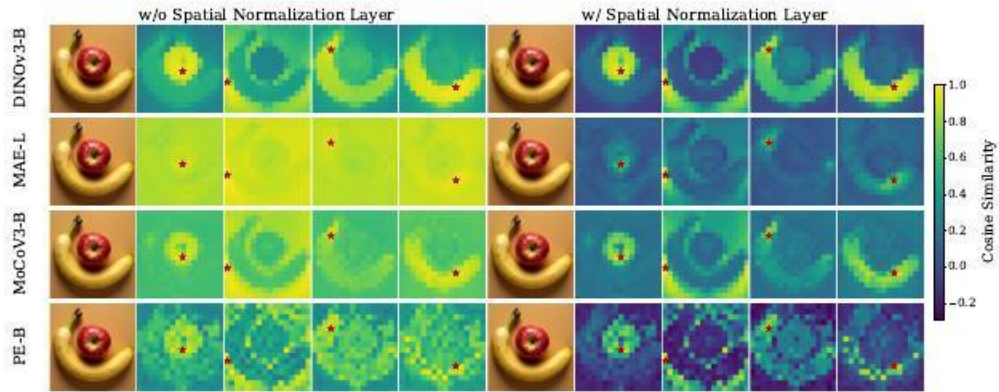


Figure 24: **Visualizing impact of spatial normalization** (Example 4). The heatmaps show token similarity patterns before and after spatial normalization. Without normalization (left), global components create high correlations across unrelated regions. With spatial normalization (right), local spatial structure is enhanced while reducing global interference, resulting in more distinct semantic boundaries.

M ADDITIONAL RESULTS

M.1 ADDITIONAL RESULTS AT HIGHER RESOLUTIONS

Setup. We conduct further experiments on ImageNet-512 (Deng et al., 2009) to evaluate the generalization of the proposed spatial improvements at higher resolutions. We follow the same setup as REPA (Yu et al., 2024), and report results across different choice of pretrained encoders (DINOv2, DINOv3, WebSSL, PE, etc.). All results are reported using SiT-XL and SiT-B using 50 NFE at inference w/o classifier free guidance. Fig. 25 shows the results with REPA before and after application of spatial improvements (iREPA). We observe that the spatial improvements also generalize to higher resolutions. Furthermore, consistent gains are observed across different choice of pretrained encoders (DINOv2, DINOv3, WebSSL, PE, etc.).

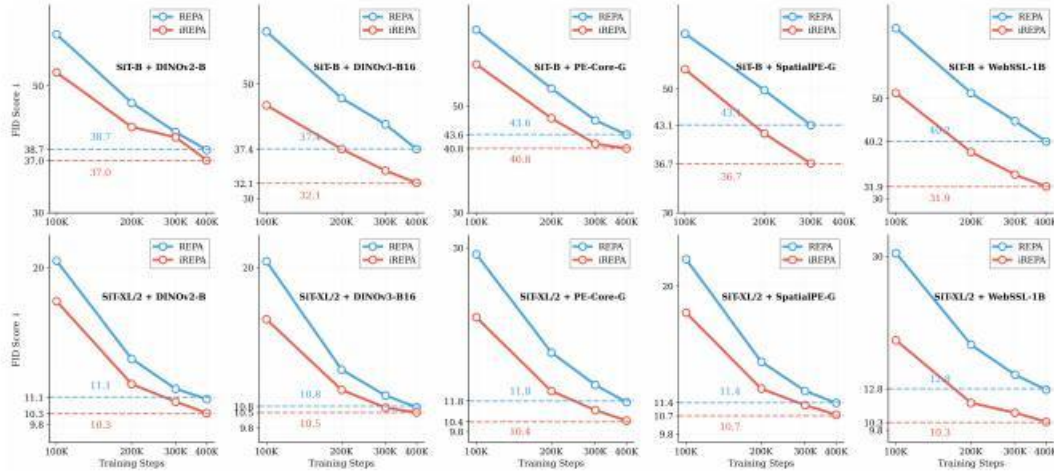


Figure 25: **Convergence results at 512x512 resolution.** Accentuating spatial features helps consistently improve convergence speed across different resolutions for both Imagenet 256 (Figure 7) and Imagenet 512 (above).

M.2 ADDITIONAL RESULTS ON MULTIMODAL T2I TASKS

Setup. To further study the generalizability of the proposed spatial improvements beyond ImageNet, we also perform extensive experiments on multimodal T2I tasks. Following REPA (Yu et al., 2024), we adopt MMDiT (Esser et al., 2024) as the diffusion backbone and apply REPA with various pretrained vision encoders (DINOv2, CLIP, WebSSL, PE, etc.). Same as REPA (Yu et al., 2024), all models are trained for 150K steps with a batch size of 256, and evaluated using an ODE sampler with 50 NFE. Fig. 26 shows the results with REPA before and after application of spatial improvements (iREPA). We observe that the spatial improvements also generalize to multimodal T2I tasks. Furthermore, consistent gains are observed across different choice of pretrained encoders (DINOv2, CLIP, WebSSL, PE, etc.).

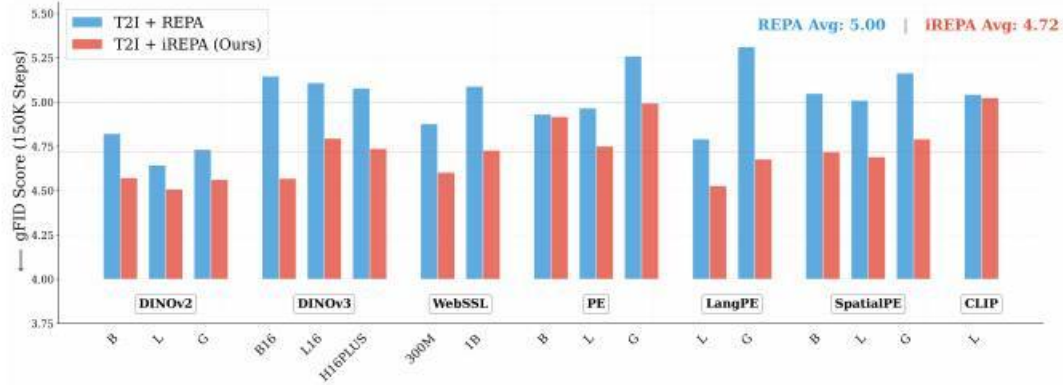


Figure 26: **Text-to-image generation across encoder variants.** Accentuating transfer of spatial features from the target representation to the diffusion features consistently improves convergence speed for both imagenet (refer §4) and multimodal T2I tasks (above). Furthermore, consistent gains are observed across different choice of pretrained encoders (DINOv2, CLIP, WebSSL, PE, etc.).

M.3 CORRELATION ANALYSIS WITHOUT OUTLIERS (MoCOv3-L AND MAE-L)

We repeat the correlation analysis from Section 3 after removing MoCOv3-L and MAE-L, which were identified as outliers. Figure 28 shows the updated correlations across different model sizes. We observe that spatial structure still shows much higher correlation with generation performance over linear probing accuracy. Interestingly, after removing the outliers linear probing actually shows a small positive correlation with gFID (i.e., as linear probing performance increases, the generation becomes worse). This trend is consistent with the observations discussed in Sec. 2, wherein often target representations with higher global semantic performance (linear probing accuracy) show similar or worse generation performance with representation alignment (REPA).

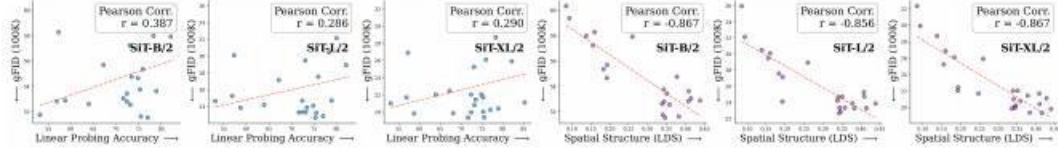


Figure 27: **Correlation analysis without outliers.** Across all model sizes (B, L, XL) spatial structure still shows much higher correlation (Pearson $|r| > 0.85$) with generation performance over linear probing accuracy (Pearson $|r| < 0.38$) after removing the outliers (MoCOv3-L and MAE-L). Interestingly, after removing the outliers linear probing actually shows a small positive correlation with gFID (i.e., as linear probing performance increases, the generation becomes worse). This trend is consistent with the observations discussed in Sec. 2, wherein often target representations with higher global semantic performance (linear probing accuracy) show similar or worse generation performance with representation alignment (REPA).

M.4 ADDITIONAL QUALITATIVE RESULTS



Figure 28: **Additional Qualitative Results** comparing generation outputs before and after application of spatial improvements with REPA. All results are reported using PE-G (Bolya et al., 2025) as the pretrained vision encoder, 400K steps (80 epochs) and with classifier-free guidance scale of 4.0. Similar to quantitative improvements (§4), we also observe that spatial improvements (iREPA) also help improve the visual quality and coherence of the generated outputs.

M.5 ADDITIONAL RESULTS WITH ALTERNATIVE EVALUATION METRICS

In addition to traditional evaluation metrics (Inception Score, FID, sFID, Precision, Recall), we also verify the robustness of the proposed findings with alternative evaluation metrics for generation quality (Stein et al., 2023; Yang et al., 2023; Kynkäänniemi et al., 2023; Jayasumana et al., 2024). In particular, we use the CMMD metric (Jayasumana et al., 2024) with the PyTorch implementation from (Paul, 2024). Standard reference set from OpenAI ADM evaluation suite (Dhariwal & Nichol, 2021) is used as reference images. We next verify the robustness of the key findings from both §3 and §4 with the CMMD metric.

Spatial structure correlates much higher with generation performance than linear probing. To study robustness of analysis from §3, we repeat the large-scale correlation analysis across different vision encoders with CMMD metric (Jayasumana et al., 2024). Results are shown in Figure 29. All results are reported using SiT-B/2 (100K steps) and REPA. All spatial metrics show much higher correlation (Pearson $|r| > 0.88$) with generation performance (CMMD) than linear probing (Pearson $|r| = 0.074$) — demonstrating that key empirical findings from §3 are robust to the choice of evaluation metric (FID or CMMD).

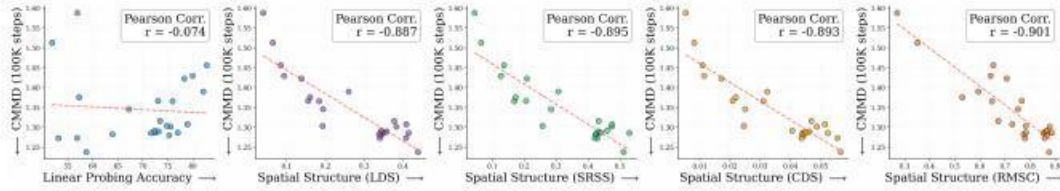


Figure 29: Correlation analysis with CMMD metric. We repeat the correlation analysis from §3 with CMMD metric (Jayasumana et al., 2024) instead of gFID. All results are reported using SiT-B/2 (100K steps) and REPA. All spatial metrics show much higher correlation (Pearson $|r| > 0.88$) with generation performance (CMMD) than linear probing (Pearson $|r| = 0.074$). This demonstrates that key empirical findings from §3 are robust to the choice of evaluation metric.

Accentuating transfer of spatial features helps improve generation quality. We next study the generalizability of analysis from §4 with the CMMD (Jayasumana et al., 2024) metric in addition to traditional evaluation metrics (gFID, sFID, IS, etc.). Results are shown in Table 8. Across various vision encoders, accentuating transfer of spatial features (iREPA) helps improve convergence speed with both CMMD and traditional evaluation metrics (IS, FID, sFID, Prec., Recall). Furthermore, consistent improvements are observed both with and without classifier-free guidance (CFG).

Vision Encoder	Steps	w/o CFG						w/ CFG					
		CMMD↓	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑	CMMD↓	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑
DINOv2-B	100K	0.702	69.20	19.3	5.89	0.64	0.61	0.529	157.2	6.35	5.91	0.77	0.54
+iREPA	100K	0.652	77.92	16.9	6.26	0.66	0.61	0.484	179.3	5.15	6.23	0.78	0.54
DINOv2-B	400K	0.455	127.4	7.76	5.06	0.70	0.66	0.320	263.0	1.98	4.60	0.80	0.61
+iREPA	400K	0.438	128.6	7.52	4.89	0.71	0.65	0.310	268.8	1.93	4.59	0.80	0.60
DINOv3-B	100K	0.749	63.64	21.4	6.14	0.63	0.60	0.571	144.0	7.57	6.09	0.76	0.53
+iREPA	100K	0.651	78.79	16.2	6.14	0.66	0.61	0.481	181.9	4.87	6.10	0.78	0.55
DINOv3-B	400K	0.474	126.7	8.10	5.06	0.70	0.66	0.336	261.2	1.99	4.58	0.80	0.61
+iREPA	400K	0.441	132.9	7.13	4.93	0.71	0.66	0.314	272.4	1.89	4.58	0.80	0.60
WebSSL-1B	100K	0.825	53.87	25.5	6.57	0.61	0.59	0.622	124.0	9.59	6.37	0.76	0.52
+iREPA	100K	0.653	77.47	16.6	6.18	0.66	0.61	0.484	177.6	5.09	6.11	0.79	0.54
WebSSL-1B	400K	0.512	116.9	9.39	5.14	0.70	0.64	0.354	250.5	2.24	4.61	0.81	0.58
+iREPA	400K	0.445	130.8	7.48	4.91	0.70	0.65	0.311	271.6	1.90	4.58	0.80	0.61
PE-Core-G	100K	0.922	42.74	32.3	6.70	0.57	0.59	0.714	97.2	14.1	6.56	0.71	0.53
+iREPA	100K	0.697	75.01	18.1	6.03	0.64	0.61	0.525	176.8	5.66	6.08	0.77	0.54
PE-Core-G	400K	0.525	109.4	10.4	5.00	0.69	0.64	0.366	238.4	2.44	4.57	0.81	0.59
+iREPA	400K	0.458	132.0	7.78	5.02	0.70	0.65	0.322	275.4	1.93	4.59	0.80	0.61
CLIP-L	100K	0.790	54.07	25.2	6.65	0.61	0.60	0.605	124.5	9.77	6.59	0.75	0.52
+iREPA	100K	0.657	74.46	17.8	6.33	0.65	0.61	0.489	172.0	5.67	6.33	0.78	0.54
CLIP-L	400K	0.487	117.8	8.97	4.98	0.70	0.65	0.339	258.7	2.15	4.64	0.81	0.59
+iREPA	400K	0.442	130.1	7.42	5.03	0.71	0.65	0.307	271.8	1.96	4.64	0.80	0.61

Table 8: **Additional results with alternative evaluation metrics.** We provide additional results with the CMMD metric (Jayasumana et al., 2024). All results are reported using SiT-XL/2, 256 batch size and traditional REPA as baseline. We adopt the pytorch implementation from (Paul, 2024) for computing the CMMD metric. Standard reference set from OpenAI ADM evaluation suite (Dhariwal & Nichol, 2021) is used as reference images. Across various vision encoders, accentuating transfer of spatial features (iREPA) helps improve convergence speed with both CMMD and traditional evaluation metrics (IS, FID etc.). This demonstrates that key empirical findings from §4 are robust to the choice of evaluation metric.

M.6 SPATIAL IMPROVEMENTS WITH SAM2

Table 9 and Figure 30 study the impact of spatial normalization (§4) on SAM2-S (46M) encoder features². As shown in Table 9, we observe that for SAM2, while spatial normalization layer helps improve performance, the improvements can be marginal. This is because spatial normalization relies on removing the global component (mean of patch tokens) to enhance spatial contrast. Since SAM2 already has little to no global information (validation accuracy < 24%), spatial normalization only slightly improves the spatial contrast (see Figure 30).

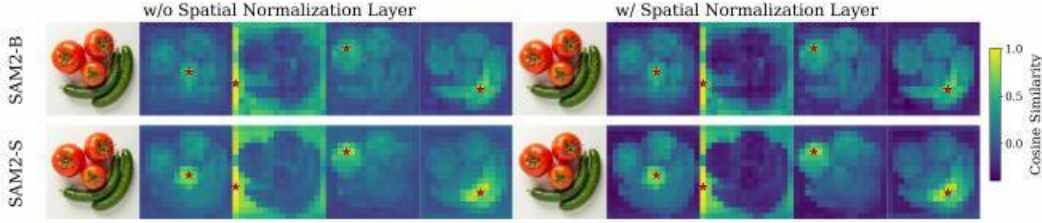


Figure 30: **Impact of spatial normalization on SAM2.** We observe that for SAM2, while use of spatial normalization layer does help enhance the spatial contrast, the improvements can be marginal. This is because spatial normalization (§4) relies on removing the global component (mean of patch tokens) to enhance spatial contrast. Since SAM2 already has little to no global information (validation accuracy < 24%), spatial normalization only slightly improves the spatial contrast.

Vision Encoder	Steps	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
SAM2-S	100K	50.69	25.32	6.52	0.631	0.588
+iREPA	100K	53.14	24.52	6.28	0.629	0.591
SAM2-S	400K	110.77	9.54	4.93	0.699	0.638
+iREPA	400K	114.62	9.10	4.89	0.704	0.640

Table 9: **Impact of spatial improvements on SAM2.** All results are reported with SiT-XL/2, without classifier-free guidance, SAM2-S (46M) as vision encoder and with traditional REPA as the baseline.

²Note that similar to (Bolya et al., 2025), we use the intermediate output of vision encoder for SAM2 features and not the mask logits. As shown in (Bolya et al., 2025), while mask logits lead to sharper spatial maps, the mask logits itself are not suitable as a target representation.

M.7 ADDITIONAL RESULTS WITH FULL-FINETUNING ACCURACY INSTEAD OF LINEAR PROBING ACCURACY

Correlation analysis with full-finetuning accuracy instead of linear probing accuracy. We repeat the correlation analysis from §3 with the validation accuracy after full-finetuning³ instead of linear probing. Results are shown in Fig. 31. All results are reported using SiT-XL/2 (100K steps) and REPA. For full-finetuning validation accuracy, instead of linear probing, we perform full-finetuning of vision encoder with learning rate of $5e-5$, warmup ratio of 0.1, and total of 30 epochs. All spatial metrics show much higher correlation with generation performance than full-finetuning accuracy. Interestingly, gFID actually shows a weak positive correlation with the validation accuracy after full-finetuning (Pearson $r = 0.317$), i.e., as validation accuracy increases, the gfid increases, and generation performance becomes worse. This is consistent with the observations discussed in §2, wherein often target representations with higher global semantic performance show similar or worse generation performance with representation alignment (REPA).

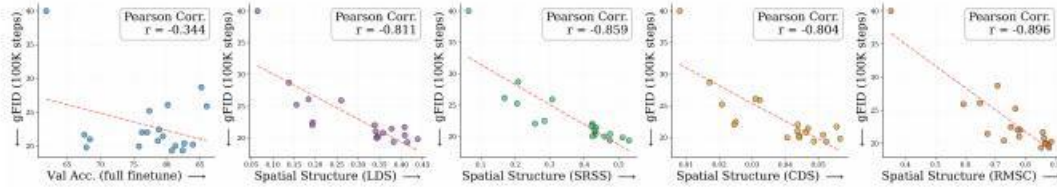


Figure 31: Correlation analysis with full-finetuning accuracy instead of linear probing accuracy. We repeat the correlation analysis from §3 with the validation accuracy after full-finetuning instead of linear probing. All results are reported using SiT-XL/2 (100K steps) and REPA. All spatial metrics show much higher correlation with generation performance than full-finetuning accuracy. Interestingly, gFID actually shows a weak positive correlation with the validation accuracy after full-finetuning (Pearson $r = 0.317$), i.e., as validation accuracy increases, the gfid increases, and generation performance becomes worse. This is consistent with the observations discussed in §2, wherein often target representations with higher global semantic performance show similar or worse generation performance with representation alignment (REPA).

³Please note that while the final findings remain similar, in context of REPA linear probing might be more accurate for estimating global information in “pretrained” vision encoders. This is because REPA uses the “pretrained” encoder features themselves for regularization. Finetuning the encoder itself can impact the amount of global information. Same as REPA (Yu et al., 2024), we therefore use linear probing accuracy as default for measuring global information.

M.8 ADDITIONAL RESULTS ON PIXEL-SPACE DIFFUSION MODELS

Method	#Params	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
PixelFlow (Chen et al., 2025)	459M	24.67	54.33	9.71	-	0.58
PixDDT (Wang et al., 2025b)	434M	36.24	46.37	17.14	-	0.63
PixNerd (Wang et al., 2025a)	458M	43.01	37.49	10.65	-	0.62
DeCo w/o $\mathcal{L}_{\text{FreqFM}}$	426M	46.44	34.12	10.41	-	0.64
DeCo + REPA (Ma et al., 2025b)	426M	48.35	31.35	9.34	-	0.65
JiT (Li & He, 2025a)	459M	29.37	49.06	11.21	0.40	0.62
JiT+iREPA (Ours)	459M	50.72	29.19	9.42	0.51	0.65

Table 10: **Generation performance of pixel-space diffusion models.** Despite its simplicity, when combined with JiT (Li & He, 2025a) iREPA outperforms recently proposed state-of-art pixel-space diffusion methods e.g., DeCo (Ma et al., 2025b). All results are reported with 200K training iterations with a batch size of 256 and evaluated using 50-step Euler sampling without classifier-free guidance. Results for DeCo are obtained directly from (Ma et al., 2025b).